LAYER: 2_MesCore

This file is a merged representation of a subset of the codebase, containing files not matching ignore patterns, combined into a single document by Repomix.

<file_summary>
This section contains a summary of this file.

<purpose>
This file contains a packed representation of a subset of the repository's contents that is considered the most important context.
It is designed to be easily consumable by AI systems for analysis, code review,
or other automated processes.
</purpose>

<file_format>
The content is organized as follows:
1. This summary section
2. Repository information
3. Directory structure
4. Repository files (if enabled)
5. Multiple file entries, each consisting of:
  - File path as an attribute
  - Full contents of the file
</file_format>

<usage_guidelines>
- This file should be treated as read-only. Any changes should be made to the
  original repository files, not this packed version.
- When processing this file, use the file path to distinguish
  between different files in the repository.
- Be aware that this file may contain sensitive information. Handle it with
  the same level of security as you would the original repository.
</usage_guidelines>

<notes>
- Some files may have been excluded based on .gitignore rules and Repomix's configuration
- Binary files are not included in this packed representation. Please refer to the Repository Structure section for a complete list of file paths, including binary files
- Files matching these patterns are excluded: **/obj/**, **/bin/**, **/*.png, **/*.jpg, **/node_modules/**, **/*.Designer.cs
- Files matching patterns in .gitignore are excluded
- Files matching default ignore patterns are excluded
- Files are sorted by Git change count (files with more changes are at the bottom)
</notes>

</file_summary>

<directory_structure>
Promatis.Net.MES.Configuration/Class1.cs
Promatis.Net.MES.Configuration/Promatis.Net.MES.Configuration.csproj
Promatis.Net.MES.Data/Configuration/TechnologicalOperationParameterBaseConfiguration.cs
Promatis.Net.MES.Data/Configuration/TechnologicalOperationUnitBaseConfiguration.cs
Promatis.Net.MES.Data/Configuration/UnitBaseConfiguration.cs
Promatis.Net.MES.Data/DBContext/MesApplicationDbContext.cs
Promatis.Net.MES.Data/Promatis.Net.MES.Data.csproj
Promatis.Net.MES.Data/Triggers/TechnologicalOperationHierarchyTrigger.cs
Promatis.Net.MES.Data/Triggers/UnitBaseHierarchyTrigger.cs
Promatis.Net.MES.Domain.Interface/Enums/CalculationMethod.cs
Promatis.Net.MES.Domain.Interface/Enums/UnitKind.cs
Promatis.Net.MES.Domain.Interface/Enums/UnitType.cs
Promatis.Net.MES.Domain.Interface/Interfaces/ITechnologicalOperation.cs
Promatis.Net.MES.Domain.Interface/Interfaces/ITechnologicalParameter.cs
Promatis.Net.MES.Domain.Interface/Promatis.Net.MES.Domain.Interface.csproj
Promatis.Net.MES.Domain/Promatis.Net.MES.Domain.csproj
Promatis.Net.MES.Domain/TechnologicalOperation/TechnologicalOperationBase.cs
Promatis.Net.MES.Domain/TechnologicalOperation/TechnologicalOperationBaseValidator.cs
Promatis.Net.MES.Domain/TechnologicalOperation/TechnologicalOperationHierarchyEngine.cs
Promatis.Net.MES.Domain/TechnologicalOperationParameter/TechnologicalOperationParameterBase.cs
Promatis.Net.MES.Domain/TechnologicalOperationParameter/TechnologicalOperationParameterBaseValidator.cs
Promatis.Net.MES.Domain/TechnologicalOperationUnit/TechnologicalOperationUnitBase.cs
Promatis.Net.MES.Domain/TechnologicalOperationUnit/TechnologicalOperationUnitBaseValidator.cs
Promatis.Net.MES.Domain/TechnologicalParameter/TechnologicalParameterBase.cs
Promatis.Net.MES.Domain/TechnologicalParameter/TechnologicalParameterBaseValidator.cs
Promatis.Net.MES.Domain/TechnologicalParameterCalcMethod/TechnologicalParameterCalcMethodBase.cs

Promatis.Net.MES.Domain/TechnologicalParameterCalcMethod/
TechnologicalParameterCalcMethodBaseValidator.cs
Promatis.Net.MES.Domain/TechnologicalParameterValue/TechnologicalParameterValueBase.cs
Promatis.Net.MES.Domain/TechnologicalParameterValue/TechnologicalParameterValueBaseValidator.cs
Promatis.Net.MES.Domain/UnitBase/UnitBase.cs
Promatis.Net.MES.Domain/UnitBase/UnitBaseValidator.cs
Promatis.Net.MES.Domain/UnitBase/UnitHierarchyEngine.cs
Promatis.Net.MES.Domain/UnitOfMeasurement/UnitOfMeasurement.cs
Promatis.Net.MES.Domain/UnitOfMeasurement/UnitOfMeasurementValidator.cs
Promatis.Net.MES.Service/Promatis.Net.MES.Service.csproj
Promatis.Net.MES.Service/TechnologicalOperationParameterService/
ITechnologicalOperationParameterService.cs
Promatis.Net.MES.Service/TechnologicalOperationParameterService/
TechnologicalOperationParameterService.cs
Promatis.Net.MES.Service/TechnologicalOperationService/ITechnologicalOperationService.cs
Promatis.Net.MES.Service/TechnologicalOperationService/TechnologicalOperationService.cs
Promatis.Net.MES.Service/TechnologicalParameterCalcMethodService/
ITechnologicalParameterCalcMethodService.cs
Promatis.Net.MES.Service/TechnologicalParameterCalcMethodService/
TechnologicalParameterCalcMethodService.cs
Promatis.Net.MES.Service/TechnologicalParameterService/ITechnologicalParameterService.cs
Promatis.Net.MES.Service/TechnologicalParameterService/TechnologicalParameterService.cs
Promatis.Net.MES.Service/UnitOfMeasurementService/IUnitOfMeasurementService.cs
Promatis.Net.MES.Service/UnitOfMeasurementService/UnitOfMeasurementService.cs
Promatis.Net.MES.Service/UnitService/IUnitBaseService.cs
Promatis.Net.MES.Service/UnitService/UnitBaseService.cs
Promatis.Net.MES.Tests/AssemblyInfo.cs
Promatis.Net.MES.Tests/Promatis.Net.MES.Tests.csproj
Promatis.Net.MES.Tests/Service/UnitServiceStandaloneTests.cs
Promatis.Net.MES.Tests/UnitBase/AppDbContext.cs
Promatis.Net.MES.Tests/UnitBase/TestUnit.cs
Promatis.Net.MES.Tests/UnitBase/TestUnitValidator.cs
Promatis.Net.MES.Tests/UnitBase/UnitBaseArchitectureTests.cs
Promatis.Net.MES.UI/_Imports.razor
Promatis.Net.MES.UI/Pages/UnitOfMeasurements/Card/UnitOfMeasurementEditDialog.razor
Promatis.Net.MES.UI/Pages/UnitOfMeasurements/Card/UnitOfMeasurementEditDialog.razor.cs
Promatis.Net.MES.UI/Pages/UnitOfMeasurements/UnitOfMeasurementPage.razor
Promatis.Net.MES.UI/Pages/UnitOfMeasurements/UnitOfMeasurementPageContext.cs
Promatis.Net.MES.UI/Promatis.Net.MES.UI.csproj
Promatis.Net.MES.UI/UiModule.cs
Promatis.Net.MES.UI/wwwroot/exampleJsInterop.js
</directory_structure>

<files>
This section contains the contents of the repository's files.

<file path="Promatis.Net.MES.Configuration/Class1.cs">
namespace Promatis.Net.MES.ConfigurationÐ
{Ð
    public class Class1Ð
    {Ð
Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Configuration/Promatis.Net.MES.Configuration.csproj">
<Project Sdk="Microsoft.NET.Sdk">Ð
Ð
  <PropertyGroup>Ð
    <TargetFramework>net10.0</TargetFramework>Ð
    <ImplicitUsings>enable</ImplicitUsings>Ð
    <Nullable>enable</Nullable>Ð
  </PropertyGroup>Ð
Ð
  <ItemGroup>Ð
    <ProjectReference Include="..\..\Core\Configuration.Web\Promatis.Net.Configuration.Web.csproj" /
>Ð
    <ProjectReference Include="..\Promatis.Net.MES.Service\Promatis.Net.MES.Service.csproj" />Ð
  </ItemGroup>Ð
Ð
</Project>
</file>

<file path="Promatis.Net.MES.Data/Configuration/
TechnologicalOperationParameterBaseConfiguration.cs">

```
using Microsoft.EntityFrameworkCore;Ð
using Microsoft.EntityFrameworkCore.Metadata.Builders;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Data;Ð
Ð
public abstract class TechnologicalOperationParameterBaseConfiguration<T, TOperation, TParameter> :
IEntityTypeConfiguration<T>Ð
    where T : TechnologicalOperationParameterBase<TOperation, TParameter> // <--- AÔ5D 5CD0CT< Cô;CäAC¤8CR ?C
    where TOperation : DomainObject, ITechnologicalOperationÐ
    where TParameter : DomainObject, ITechnologicalParameterÐ
{Ð
    public virtual void Configure(EntityTypeBuilder<T> builder)Ð
    {Ð
        // 1. B4=C„:C ;DÄ=D´9 C„=CD5C¤A CD;Dò ?D 5CD>D$2D 0D”5CÔ8Dò 4D41C´8D >C$0CÔ8Dò ?C @C <CTBD >C" =C  >C
        builder.HasIndex(x => new { x.OperationId, x.ParameterId })Ð
            .IsUnique();Ð
Ð
        // 2. B  AÔ%B  AÔ At Bd Bò  TU%’ d”ÅDU" „ D ?D 0C$;CT=C„5 CD;Dò 6ögB FVÆWFR•
        // A„AC¤;DäGC 5CÂ 7C ?C„ADÂ 8Cr 2D´1Cä@C¤8, CTAC´8 D44C ;CT=C  AC <C  AC$OCtL (ISoftDeletable.Deleted
        // A„ A, 5D ;C, 1D´; CÄOC4:Câ CCD0C´5CÒ AC < Cô@C„2Dô7C =CÔKC’ BCTECÔ>C´>C48Dt5D :C„9 Cô0D 0CÄ5D$@ (P
        builder.HasQueryFilter(x =>Ð
            x.DeletedAt == null &&Ð
            EF.Property<DateTime?>(x.Parameter, "DeletedAt") == null);Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Data/Configuration/TechnologicalOperationUnitBaseConfiguration.cs">
using Microsoft.EntityFrameworkCore;Ð
using Microsoft.EntityFrameworkCore.Metadata.Builders;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Data;Ð
Ð
/// <summary>Ð
/// A¤>CÔDC„3D4@C FC„O CD;Dò 1C 7D² AC$OCt8Ð
/// </summary>Ð
/// <typeparam name="T"></typeparam>Ð
public abstract class TechnologicalOperationUnitBaseConfiguration<T, TOperation> :
IEntityTypeConfiguration<T>Ð
    where T : TechnologicalOperationUnitBase<TOperation> // <--- AÔ5D 5CD0CT< D$8Cò >Cô5D 0Dd8C•
    where TOperation : DomainObject, ITechnologicalOperationÐ
{Ð
    public virtual void Configure(EntityTypeBuilder<T> builder)Ð
    {Ð
        // 1. B4=C„:C ;DÄ=D´9 C„=CD5C¤A CD;Dò ?C @D² ?Cä;CT9 (AÔ@CT4CäBC$@C IC 5D" 4D41C´8D >C$0CÔ8CR AC$OCt5
        builder.HasIndex(x => new { x.OperationId, x.UnitId })Ð
            .IsUnique();Ð
Ð
        // 2. B  AÔ%B  AÔ At Bd Bò  TU%’ d”ÅDU" „ D ?D 0C$;CT=C„5 CD;Dò 6ögB FVÆWFR•
        // BD8C´LD$@D45CÂ AC <D2 AC$OCtCDäID4N D CD”=CäAD$L, CTAC´8 Cä=C  ?Cä<CTGCT=C  :C : D44C ;CT=CÔ0DòÍ
        // A„ A, 5D ;C, CCD0C´5CÔ> D 2Dô7C =CÔ>CR A CÔ5C’ >C >D CCD>C$0CÔ8CR …Væ—B’í
        // A„ACô>C´LCtCCT< EF.Property CD;Dò 1CT7Cä?C ACÔ>C4> CD>D BD4?C  : D$5CÔ5C$KCÂôOC$=D´< D 2Cä9D BC$0C
        builder.HasQueryFilter(x =>Ð
            x.DeletedAt == null &&Ð
            EF.Property<DateTime?>(x.Unit, "DeletedAt") == null);Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Data/Configuration/UnitBaseConfiguration.cs">
using Microsoft.EntityFrameworkCore;Ð
using Microsoft.EntityFrameworkCore.Metadata.Builders;Ð
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.Data;Ð
Ð
public class UnitBaseConfiguration : IEntityTypeConfiguration<UnitBase>Ð
{Ð
    public void Configure(EntityTypeBuilder<UnitBase> builder)Ð
    {Ð
```

```
        // A$:C´NDt0CT< D BD 0D$5C48Dâ E B 4C´O C$ACT9 C„5D 0D EC„8 UnitBaseÐ
        builder.UseTptMappingStrategy();Ð
Ð
        // AÔ0D BD >C”:C  ?Cä;CT9Ð
        builder.Property(x => x.Kind).IsRequired();Ð
        builder.Property(x => x.Type).IsRequired();Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Data/DBContext/MesApplicationDbContext.cs">
using Microsoft.EntityFrameworkCore;Ð
using Microsoft.Extensions.Configuration;Ð
using Promatis.Net.Data;Ð
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.Data;Ð
Ð
public class MesApplicationDbContext(Ð
    DbContextOptions options,Ð
    IConfiguration configuration,Ð
    IServiceProvider? serviceProvider = null) Ð
    : ApplicationDbContext(options, configuration, serviceProvider)Ð
{Ð
    public DbSet<UnitOfMeasurement> UnitOfMeasurements => Set<UnitOfMeasurement>();Ð
}
</file>

<file path="Promatis.Net.MES.Data/Promatis.Net.MES.Data.csproj">
<Project Sdk="Microsoft.NET.Sdk">Ð
Ð
  <PropertyGroup>Ð
    <TargetFramework>net10.0</TargetFramework>Ð
    <ImplicitUsings>enable</ImplicitUsings>Ð
    <Nullable>enable</Nullable>Ð
  </PropertyGroup>Ð
Ð
  <ItemGroup>Ð
    <PackageReference Include="Microsoft.EntityFrameworkCore" Version="10.0.8" />Ð
    <PackageReference Include="Microsoft.EntityFrameworkCore.Relational" Version="10.0.8" />Ð
    <PackageReference Include="Microsoft.Extensions.Configuration" Version="10.0.8" />Ð
  </ItemGroup>Ð
Ð
  <ItemGroup>Ð
    <ProjectReference Include="..\..\Core\Data\Promatis.Net.Data.csproj" />Ð
    <ProjectReference Include="..\Promatis.Net.MES.Domain\Promatis.Net.MES.Domain.csproj" />Ð
  </ItemGroup>Ð
Ð
</Project>
</file>

<file path="Promatis.Net.MES.Data/Triggers/TechnologicalOperationHierarchyTrigger.cs">
using Microsoft.EntityFrameworkCore;Ð
using Promatis.Net.Data;Ð
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.Data;Ð
Ð
/// <summary>Ð
/// A 1D BD 0C¤BCÔKC’ 1C 7Cä2D´9 D$@C„3C45D  8CT@C @DT8C, BCTECÔ>C´>C48Dt5D :C„E Cä?CT@C FC„9.Ð
/// At0D”8D”0CTB D BD CC¤BD4@D2 >D" =CT:Cä@D 5C¤BCÔ>C4> CD>C 0C$;CT=C„O Cô>CDCCt;Cä2 C, :Cä=D$@Cä;C„@D45D" AD
/// </summary>Ð
public abstract class TechnologicalOperationHierarchyTrigger<T, TOLink, TPLink> :
IBeforeSaveTrigger<T>Ð
    where T : TechnologicalOperationBase<T, TOLink, TPLink>Ð
    where TOLink : classÐ
    where TPLink : classÐ
{Ð
    public async Task HandleAsync(EntityCancelEventArgs<T> args)Ð
    {Ð
        T operation = args.Entity;Ð
Ð
        // ATAC´8 Cä?CT@C FC„O C¤>D =CT2C O — Cô@Cä?D4AC¤0CT< Cô@Cä2CT@C¤CÐ
        if (!operation.ParentId.HasValue || operation.ParentId == Guid.Empty) return;Ð
Ð
        // A„7C$;CT:C 5CÂ 8Cr !B4 AB AD$0D$CD  —4ÆV b 8 Name D >CD8D$5C´LD :Cä9 Cä?CT@C FC„8Ð
```

```
        var parentInfo = await args.Context.Set<T>()Ð
            .Where(x => x.Id == operation.ParentId)Ð
            .Select(x => new { x.IsLeaf, x.Name })Ð
            .FirstOrDefaultAsync();Ð
Ð
        if (parentInfo != null)Ð
        {Ð
            // A„ACô>C´LCtCCT< CÔ0D, 4Cä<CT=CÔKC’ 4C$8Cd>C¢ 4C´O Cô@Cä2CT@C¤8 Cô@C 2C„; C$;Cä6CT=CÔ>D BC,í
            // Aô5D 5CD0CT< D BC BD4A IsLeaf D >CD8D$5C´O, C$KD$0D”5CÔ=D´9 C„7 C 0CtK.Ð
            if (parentInfo.IsLeaf)Ð
            {Ð
                args.Cancel = true;Ð
                args.ErrorMessage = $"AÔ0D CD„5CÔ8CR 8CT@C @DT8C, ÔU3¢ "CTECÔ>C´>C48Dt5D :C O Cä?CT@C FC„O '{
                                    $"CÔ5 CÄ>Cd5D" 1D´BDÂ 2C´>Cd5CÔ0 C" w·  &VçD–æfòäæ ÖWÒrÂ BC : C¤0C¢ >CÔ0
                return;Ð
            }Ð
        }Ð
Ð
        // AD>Cô>C´=C„BCT;DÄ=C O Ct0D”8D$0 CÔ0 C„7CÄ5CÔ5CÔ8CS¢ 5D ;C, >Cô5D 0Dd8Dò AD$0CÔ>C$8D$ADò  C„AD$>CÂÂ
        // Cô@Cä2CT@Dô5CÂ 2 ChangeTracker C„;C,  ABÂ =CTB C´8 D2 =CT5 CD>Dt5D =C„E DÔ;CT<CT=D$>C-
        if (operation.IsLeaf)Ð
        {Ð
            bool hasChildrenInDb = await args.Context.Set<T>()Ð
                .AnyAsync(x => x.ParentId == operation.Id);Ð
Ð
            if (!TechnologicalOperationHierarchyEngine.CanChangeLeafStatus(hasChildrenInDb,
operation.IsLeaf))Ð
            {Ð
                args.Cancel = true;Ð
                args.ErrorMessage = $"AÔ0D CD„5CÔ8CR FCT;CäAD$=CäAD$8 CD0CÔ=D´E: AÔ5C$>Ct<Cä6CÔ> D 4CT;C BDÂ
                                    $"D$0C¢ :C : C$=D4BD 8 CÔ5CR CCd5 D >CD5D 6C BD O CD@D43C„5 D$5DT=Cä;Cä3C
            }Ð
        }Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Data/Triggers/UnitBaseHierarchyTrigger.cs">
using Microsoft.EntityFrameworkCore;Ð
using Promatis.Net.Data;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Data;Ð
Ð
/// <summary>Ð
/// B$@C„3C45D  4C´O Cô@Cä2CT@C¤8 D ?CTFC„DC„GCÔKDR ?D 0C$8C² 8CT@C @DT8C, Væ–D& 6Rí
/// </summary>Ð
public class UnitBaseHierarchyTrigger : IBeforeSaveTrigger<UnitBase>Ð
{Ð
    public async Task HandleAsync(EntityCancelEventArgs<UnitBase> args)Ð
    {Ð
        UnitBase unit = args.Entity;Ð
Ð
        if (!unit.ParentId.HasValue || unit.ParentId == Guid.Empty) return;Ð
Ð
        // Aô>C´CDt0CT< Kind C, æ ÖR @Cä4C„BCT;Dò 4C´O C„=DD>D <C BC„2CÔ>C4> D >Cä1D”5CÔ8Dý
        var parentInfo = await args.Context.Set<UnitBase>()Ð
            .Where(x => x.Id == unit.ParentId)Ð
            .Select(x => new { x.Kind, x.Name })Ð
            .FirstOrDefaultAsync();Ð
Ð
        if (parentInfo != null && !UnitHierarchyEngine.IsHierarchyValid(parentInfo.Kind, unit.Kind))Ð
        {Ð
            args.Cancel = true;Ð
            args.ErrorMessage = $"AÔ0D CD„5CÔ8CR 8CT@C @DT8C, ÔU3¢  C JCT:D" w·Væ–Bäæ ÖWÒr ‡·Væ–Bä¶–æGÒ’ " ½
                                $"CÔ5 CÄ>Cd5D" 1D´BDÂ 2C´>Cd5CÒ 2 '{parentInfo.Name}' ({parentInfo.Kind}).";Ð
        }Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Domain.Interface/Enums/CalculationMethod.cs">
using System.ComponentModel;Ð
Ð
namespace Promatis.Net.MES.Domain.Interface;Ð
```

```
Ð
/// <summary>Ð
/// AÄ5D$>CDK D 0D GCTBC  BCTE. Cô0D 0CÄ5D$@Cä2Ð
/// </summary>Ð
[Flags]Ð
public enum CalculationMethodÐ
{Ð
    [Description("AÔ5 Ct0CD0CÒ"•Ý
    None = 0,               // 00000000Ð
Ð
    [Description("AÄ0C¤AC„<C ;DÄ=Cä5 Ct=C GCT=C„5 (MAX)")]Ð
    Max = 1,Ð
Ð
    [Description("AÄ8CÔ8CÄ0C´LCÔ>CR 7CÔ0Dt5CÔ8CR „Ô”â’"•Ý
    Min = 2,Ð
Ð
    [Description("B @CT4CÔ5CR 7CÔ0Dt5CÔ8CR „ dr’"•Ý
    Avg = 4,Ð
Ð
    [Description("AôD 2Cä5 Ct=C GCT=C„5 (FIRST)")]Ð
    First = 8,Ð
Ð
    [Description("Aô>D ;CT4CÔ5CR 7CÔ0Dt5CÔ8CR „Ä 5B’"•Ý
    Last = 16,Ð
Ð
    [Description("A$ACR 7CÔ0Dt5CÔ8Dò „ ÄÂ’"•Ý
    All = 32Ð
}
</file>

<file path="Promatis.Net.MES.Domain.Interface/Enums/UnitKind.cs">
using System.ComponentModel;Ð
Ð
namespace Promatis.Net.MES.Domain.Interface;Ð
Ð
public enum UnitKindÐ
{Ð
    [Description("B :C´0CDAC¤0Dò ;Cä3C„AD$8C¤0")]Ð
    Storage = UnitType.Storage | UnitType.Zone | UnitType.Rack | UnitType.Cell | UnitType.Crane,Ð
Ð
    [Description("Aô@Cä8Ct2Cä4D BC$5CÔ=C O Ct>CÔ0")]Ð
    Production = UnitType.Workshop | UnitType.Section | UnitType.Line | UnitType.Workstation |
UnitType.MachineTool | UnitType.Table,Ð
Ð
    [Description("B$@C =D ?Cä@D$=D´9 D47CT;")]Ð
    Transport = UnitType.Vehicle | UnitType.Conveyor,Ð
Ð
    [Description("Aô>CD@C 7CD5C´5CÔ8CR"•Ý
    Department = UnitType.Workshop | UnitType.Section | UnitType.Other,Ð
Ð
    [Description("B 0C >Dt0Dò BCäGC¤0 / BôGCT9C¤0")]Ð
    Position = UnitType.Cell | UnitType.OtherÐ
}
</file>

<file path="Promatis.Net.MES.Domain.Interface/Enums/UnitType.cs">
using System.ComponentModel;Ð
Ð
namespace Promatis.Net.MES.Domain.Interface;Ð
Ð
[Flags]Ð
public enum UnitTypeÐ
{Ð
    None = 0,Ð
Ð
    [Description("Bd5DR"•Ý
    Workshop = 1,Ð
Ð
    [Description("B4GC AD$>C¢"•Ý
    Section = 2,Ð
Ð
    [Description("A´8CÔ8Dò ò  Cä=C$5C”5D "•Ý
    Line = 4,Ð
Ð
    [Description("B 0C >Dt5CR <CTAD$>")]Ð
    Workstation = 8,Ð
```

```
Ð
    [Description("B :C´0CB"•Ý
    Storage = 16,Ð
Ð
    [Description("At>CÔ0 DT@C =CT=C„O")]Ð
    Zone = 32,Ð
Ð
    [Description("B BCT;C´0Cb"•Ý
    Rack = 64,Ð
Ð
    [Description("BôGCT9C¤0 C 4D 5D 0 DT@C =CT=C„O")]Ð
    Cell = 128,Ð
Ð
    [Description("A¤@C = / Aô>CDJCT<CÔ8C¢"•Ý
    Crane = 256,Ð
Ð
    [Description("B BC =Cä:")]Ð
    MachineTool = 512,Ð
Ð
    [Description("A$5D AD$0C¢ ò !D$>C²"•Ý
    Table = 1024,Ð
Ð
    [Description("B$@C =D ?Cä@D$=Cä5 D @CT4D BC$>")]Ð
    Vehicle = 2048,Ð
Ð
    [Description("A 2D$>CÔ>CÄ=D´9 D$@C =D ?Cä@D$5D "•Ý
    Conveyor = 4096,Ð
Ð
    [Description("Aô@CäGCT5")]Ð
    Other = 8192Ð
}
</file>

<file path="Promatis.Net.MES.Domain.Interface/Interfaces/ITechnologicalOperation.cs">
namespace Promatis.Net.MES.Domain.Interface;Ð
Ð
public interface ITechnologicalOperationÐ
{Ð
    Ð
}
</file>

<file path="Promatis.Net.MES.Domain.Interface/Interfaces/ITechnologicalParameter.cs">
namespace Promatis.Net.MES.Domain.Interface;Ð
Ð
public interface ITechnologicalParameterÐ
{Ð
    Ð
}
</file>

<file path="Promatis.Net.MES.Domain.Interface/Promatis.Net.MES.Domain.Interface.csproj">
<Project Sdk="Microsoft.NET.Sdk">Ð
Ð
  <PropertyGroup>Ð
    <TargetFramework>net10.0</TargetFramework>Ð
    <ImplicitUsings>enable</ImplicitUsings>Ð
    <Nullable>enable</Nullable>Ð
  </PropertyGroup>Ð
Ð
  <ItemGroup>Ð
    <ProjectReference Include="..\..\Core\Domain.Interface\Promatis.Net.Domain.Interface.csproj" />Ð
  </ItemGroup>Ð
Ð
</Project>
</file>

<file path="Promatis.Net.MES.Domain/Promatis.Net.MES.Domain.csproj">
<Project Sdk="Microsoft.NET.Sdk">Ð
Ð
  <PropertyGroup>Ð
    <TargetFramework>net10.0</TargetFramework>Ð
    <ImplicitUsings>enable</ImplicitUsings>Ð
    <Nullable>enable</Nullable>Ð
  </PropertyGroup>Ð
Ð
```

```
  <ItemGroup>Ð
    <ProjectReference Include="..\..\Core\Domain\Promatis.Net.Domain.csproj" />Ð
    <ProjectReference Include="..
\Promatis.Net.MES.Domain.Interface\Promatis.Net.MES.Domain.Interface.csproj" />Ð
  </ItemGroup>Ð
Ð
</Project>
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalOperation/TechnologicalOperationBase.cs">
using Promatis.Net.Domain;Ð
using Promatis.Net.Domain.Interface;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
/// <summary>Ð
/// A 0Ct>C$0Dò BCTECÔ>C´>C48Dt5D :C O Cä?CT@C FC„O.Ð
/// AÔ0D ;CT4D45D" GC„AD$>CR =CRÔ4Cd5CÔ5D 8C¢ 4CT@CT2Câ !B4 ABÂ CC 8D 0Dò :Cä=DD;C„:D$K CÄ0Cô?C„=C40,Ð
/// C, ?D 5CD>D BC 2C´OCTB D BD >C4> D$8Cô8Ct8D >C$0CÔ=D´9 C„=D$5D DCT9D  4C´O C 8Ct=CTA-C´>C48C¤8.Ð
/// </summary>Ð
public abstract class TechnologicalOperationBase<T, TOLink, TPLink> : ReferenceTreeBase<T>, // <- A¤ A„"A„'AT
    ITechnologicalOperationÐ
    where T : TechnologicalOperationBase<T, TOLink, TPLink>Ð
    where TOLink : classÐ
    where TPLink : classÐ
{Ð
    public bool IsLeaf { get; set; } = true;Ð
Ð
    public virtual ICollection<TOLink> UnitLinks { get; set; } = new List<TOLink>();Ð
    public virtual ICollection<TPLink> ParameterLinks { get; set; } = new List<TPLink>();Ð
}
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalOperation/TechnologicalOperationBaseValidator.cs">
using FluentValidation;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.Domain.Interface;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
// A$0C´8CD0D$>D  >Cô5D 0Dd8C, „>D BC 5D$ADò 1CT7 C„7CÄ5CÔ5CÔ8C'•
public abstract class TechnologicalOperationBaseValidator<T> : ReferenceTreeBaseValidator<T>Ð
    where T : ReferenceTreeBase<T>, ITreeNode<T>, ITechnologicalOperation // <- A¤ A„"A„'AT!A¤ AR  B  B  A$ A
{Ð
    protected TechnologicalOperationBaseValidator() : base()Ð
    {Ð
        // A 0Ct>C$KC' &VfW&Væ6T& 6Uf Æ–F F÷" CCd5 Cô@Cä2CT@C„; Cä1D”8CR ?Cä;Dò „–BÂ æ ÖR•
        // At4CTADÂ <D² ?C„HCT< Cô@C 2C„;C  AD$@Cä3Câ 4C´O D$5DT=Cä;Cä3C„GCTAC¤8DR >Cô5D 0Dd8C•
Ð
        RuleFor(x => x.Code)Ð
            .NotEmpty()Ð
            .WithMessage("A¤>CB >Cô5D 0Dd8C, >C OCt0D$5C´5CÒâ"•
            .MaximumLength(50)Ð
            .WithMessage("A¤>CB >Cô5D 0Dd8C, =CR 4Cä;Cd5CÒ ?D 5C$KD„0D$L 50 D 8CÄ2Cä;Cä2.");Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalOperation/
TechnologicalOperationHierarchyEngine.cs">
using System.Reflection;Ð
using Promatis.Net.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
public static class TechnologicalOperationHierarchyEngineÐ
{Ð
    /// <summary>Ð
    /// A4;Cä1C ;DÄ=D´9 C„AD$>Dt=C„: Cô@C 2CDK: Cô@Cä2CT@Dô5D"Â 2C ;C„4CÔ> C´8 C$;Cä6CT=C„5 CD>Dt5D =CT9 Cä?C
    /// AÔ0CÄ5D BC$> C ;Cä:C„@D45D" ACä7CD0CÔ8CR 2CTBCä: C$=D4BD 8 D$5D <C„=C ;DÄ=D´E Cä?CT@C FC„9 (IsLeaf =
    /// B$8Cô8Ct0Dd8Dò ?CT@CT2CT4CT=C  =C  GC„AD$KC' :Cä=D$@C :D" 4CT@CT2C  BÂ GD$> Cô>C´=CäAD$LDâ CD BD 0CÔO
    /// </summary>Ð
    /// <typeparam name="T">A¤>CÔ:D 5D$=D´9 D$8Cò BCTECÔ>C´>C48Dt5D :Cä9 Cä?CT@C FC„8</typeparam>Ð
    /// <param name="parent">B >CD8D$5C´LD :C O Cä?CT@C FC„O</param>Ð
```

```
    /// <param name="childIsLeaf">Aô@C„7CÔ0C¢ BCä3CâÂ OC$;Dô5D$ADò ;C, ACä7CD0C$0CT<D´9/Cô5D 5CÄ5D”0CT<D´9 D
    public static bool IsHierarchyValid<T>(T? parent, bool childIsLeaf)Ð
        where T : class, ITreeNode<T>Ð
    {Ð
        // 1. ATAC´8 D >CD8D$5C´O CÔ5D"Â <D² ACä7CD0CT< C¤>D =CT2Cä9 DÔ;CT<CT=D" r MD$> C$ACT3CD0 C$0C´8CD=Câ
        if (parent == null)Ð
        {Ð
            return true;Ð
        }Ð
Ð
        // 2. Bt5D 5Cr @CTDC´5C¤AC„N (C„;C, :C AD" : C„=D$5D DCT9D C, CTAC´8 C K Cä= C KC²’ 8Ct2C´5C¤0CT< Cô@
        // B$0C¢ :C : T CÔ0D ;CT4D45D$ADò >D" FV6†æöÆöv–6 Ä÷ W& F–öä& 6RÂ AC$>C”AD$2Câ —4ÆV b BC < C40D 0CÔBC
        PropertyInfo? isLeafProperty = typeof(T).GetProperty("IsLeaf");Ð
        if (isLeafProperty != null)Ð
        {Ð
            bool parentIsLeaf = (bool)isLeafProperty.GetValue(parent)!;Ð
            if (parentIsLeaf)Ð
            {Ð
                return false; // B$5D <C„=C ;DÄ=D´9 D47CT; CÔ5 CÄ>Cd5D" 1D´BDÂ @Cä4C„BCT;CT<Ð
            }Ð
        }Ð
Ð
        return true;Ð
    }Ð
Ð
    /// <summary>Ð
    /// Aô@Cä2CT@Dô5D"Â @C 7D 5D„5CÔ> C´8 Cô5D 5C¤;DäGC BDÂ DC´0C2 —4ÆV b C D CD”5D BC$CDäICT9 Cä?CT@C FC„8.Ð
    /// ATAC´8 D2 >Cô5D 0Dd8C, CCd5 DD8Ct8Dt5D :C, 5D BDÂ 4CäGCT@CÔ8CR 2CTBC¤8 C"  At#, Cô@CT2D 0D”0D$L CTQ C
    /// </summary>Ð
    public static bool CanChangeLeafStatus(bool hasChildren, bool requestedIsLeaf)Ð
    {Ð
        if (hasChildren && requestedIsLeaf)Ð
        {Ð
            return false;Ð
        }Ð
Ð
        return true;Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalOperationParameter/
TechnologicalOperationParameterBase.cs">
using Promatis.Net.Domain;Ð
using Promatis.Net.Domain.Interface;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
/// <summary>Ð
/// B 2Dô7D4ND”0Dò AD4ICÔ>D BDÂ ´ CÔ>C48CRÔ:CâÔ<CÔ>C48CË² <CT6CDC D$5DT=Cä;Cä3C„GCTAC¤8CÄ8 Cä?CT@C FC„OCÄ8 C,
/// Aä?D 5CD5C´OCTB D$@CT1Cä2C =C„O C¢ 7C ?Cä;CÔ5CÔ8Dâ ?C @C <CTBD >C" 4C´O C¤>CÔ:D 5D$=Cä9 Cä?CT@C FC„8.Ð
/// </summary>Ð
public abstract class TechnologicalOperationParameterBase<TOperation, TParameter> : DomainObject,
ISoftDeletableÐ
    where TOperation : DomainObject, ITechnologicalOperationÐ
    where TParameter : DomainObject, ITechnologicalParameterÐ
{Ð
    public Guid OperationId { get; set; }Ð
Ð
    /// <summary>Ð
    /// AÔ0C$8C40Dd8Cä=CÔ>CR AC$>C”AD$2Câ : C 0Ct>C$>C’ >Cô5D 0Dd8C,í
    /// </summary>Ð
    public virtual TOperation Operation { get; set; } = null!;Ð
Ð
    public Guid ParameterId { get; set; }Ð
Ð
    /// <summary>Ð
    /// AÔ0C$8C40Dd8Cä=CÔ>CR AC$>C”AD$2Câ : C 0Ct>C$>CÄC Cô0D 0CÄ5D$@D2í
    /// </summary>Ð
    public virtual TParameter Parameter { get; set; } = null!;Ð
Ð
    /// <summary>Ð
    /// Aô@C„7CÔ0C¢ BCä3CâÂ GD$> Cô0D 0CÄ5D$@ Cä1Dô7C BCT;CT= CD;Dò 7C ?Cä;CÔ5CÔ8Dòí
    /// </summary>Ð
    public bool IsRequired { get; set; } = false;Ð
```

```
Ð
    /// <summary>Ð
    /// AÔ>CÄ8CÔ0C´LCÔ>CR 7CÔ0Dt5CÔ8CR ?C @C <CTBD 0 Cô> D4<Cä;Dt0CÔ8Dâí
    /// </summary>Ð
    public string? DefaultValue { get; set; }Ð
Ð
    /// <summary>Ð
    /// AÄ8CÔ8CÄ0C´LCÔ> CD>CôCD BC„<Cä5 Ct=C GCT=C„5.Ð
    /// </summary>Ð
    public double? MinValue { get; set; }Ð
Ð
    /// <summary>Ð
    /// AÄ0C¤AC„<C ;DÄ=Câ 4Cä?D4AD$8CÄ>CR 7CÔ0Dt5CÔ8CRí
    /// </summary>Ð
    public double? MaxValue { get; set; }Ð
Ð
    public DateTime? DeletedAt { get; set; }Ð
    public string? DeletedBy { get; set; }Ð
}
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalOperationParameter/
TechnologicalOperationParameterBaseValidator.cs">
using FluentValidation;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
public abstract class TechnologicalOperationParameterBaseValidator<T, TOperation, TParameter> :
DomainObjectValidator<T>Ð
    where T : TechnologicalOperationParameterBase<TOperation, TParameter>Ð
    where TOperation : DomainObject, ITechnologicalOperationÐ
    where TParameter : TechnologicalParameterBaseÐ
{Ð
    protected TechnologicalOperationParameterBaseValidator()Ð
    {Ð
        RuleFor(x => x.OperationId)Ð
            .NotEmpty()Ð
            .WithMessage("A„4CT=D$8DD8C¤0D$>D  >Cô5D 0Dd8C, >C OCt0D$5C´5CÒâ"“½
Ð
        RuleFor(x => x.ParameterId)Ð
            .NotEmpty()Ð
            .WithMessage("A„4CT=D$8DD8C¤0D$>D  BCTECÔ>C´>C48Dt5D :Cä3Câ ?C @C <CTBD 0 Cä1Dô7C BCT;CT=.");Ð
Ð
        // A¤@CäAD Ô?Cä;CT2C O C$0C´8CD0Dd8Dó¢  C :D 8CÄ0C´LCÔ>CR 7CÔ0Dt5CÔ8CR =CR <Cä6CTB C KD$L CÄ5CÔLD„5 C
        RuleFor(x => x.MaxValue)Ð
            .GreaterThanOrEqualTo(x => x.MinValue)Ð
            .When(x => x.MinValue.HasValue && x.MaxValue.HasValue)Ð
            .WithMessage("AÄ0C¤AC„<C ;DÄ=Câ 4Cä?D4AD$8CÄ>CR 7CÔ0Dt5CÔ8CR =CR <Cä6CTB C KD$L CÄ5CÔLD„5 CÄ8CÔ8C
    }Ð
}
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalOperationUnit/TechnologicalOperationUnitBase.cs">
using Promatis.Net.Domain;Ð
using Promatis.Net.Domain.Interface;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
/// <summary>Ð
/// A 1D BD 0C¤BCÔ0Dò 1C 7C  4C´O D 2Dô7C, „@C AD„8D OCT<C O)Ð
/// </summary>Ð
public abstract class TechnologicalOperationUnitBase<TOperation> : DomainObject, ISoftDeletableÐ
    where TOperation : DomainObject, ITechnologicalOperationÐ
{Ð
    public Guid OperationId { get; set; }Ð
Ð
    /// <summary>Ð
    /// AÔ0C$8C40Dd8Cä=CÔ>CR AC$>C”AD$2Câ : Cä?CT@C FC„8 (C CCD5D" 7C :D KD$> C¤>CÔ:D 5D$=D´< D$8Cô>CÂ 2 MDM)
    /// </summary>Ð
    public virtual TOperation Operation { get; set; } = null!;Ð
Ð
    public Guid UnitId { get; set; }Ð
Ð
```

```
    /// <summary>Ð
    /// Aô@Dô<C O D AD´;C¤0 CÔ0 C 0Ct>C$KC’ :C´0D A Cä1Cä@D44Cä2C =C„O. Ð
    /// EF Core C 2D$>CÄ0D$8Dt5D :C, AC$OCd5D" MD$> D  BC 1C´8Dd5C’ >C >D CCD>C$0CÔ8Dò ?Câ ACä3C´0D„5CÔ8Dô<.Ð
    /// </summary>Ð
    public virtual UnitBase Unit { get; set; } = null!;Ð
Ð
    public int Priority { get; set; } = 1;Ð
Ð
    public DateTime? DeletedAt { get; set; }Ð
    public string? DeletedBy { get; set; }Ð
}
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalOperationUnit/
TechnologicalOperationUnitBaseValidator.cs">
using FluentValidation;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
/// <summary>Ð
/// A 0Ct>C$KC’ 2C ;C„4C BCä@ D 2Dô7C•
/// </summary>Ð
public abstract class TechnologicalOperationUnitBaseValidator<T, TOperation> :
DomainObjectValidator<T>Ð
    where T : TechnologicalOperationUnitBase<TOperation>Ð
    where TOperation : DomainObject, ITechnologicalOperationÐ
{Ð
    protected TechnologicalOperationUnitBaseValidator()Ð
    {Ð
        RuleFor(x => x.OperationId)Ð
            .NotEmpty()Ð
            .WithMessage("A„4CT=D$8DD8C¤0D$>D  >Cô5D 0Dd8C, >C OCt0D$5C´5CÒâ"“½
Ð
        RuleFor(x => x.UnitId)Ð
            .NotEmpty()Ð
            .WithMessage("A„4CT=D$8DD8C¤0D$>D  >C >D CCD>C$0CÔ8Dò >C OCt0D$5C´5CÒâ"“½
Ð
        RuleFor(x => x.Priority)Ð
            .InclusiveBetween(1, 10)Ð
            .WithMessage("Aô@C„>D 8D$5D" >C >D CCD>C$0CÔ8Dò 4Cä;Cd5CÒ 1D´BDÂ 2 CD8C ?C 7Cä=CR >D"   4Câ   â"“
    }Ð
}
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalParameter/TechnologicalParameterBase.cs">
using Promatis.Net.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
/// <summary>Ð
/// A 0Ct>C$KC’ BCTECÔ>C´>C48Dt5D :C„9 Cô0D 0CÄ5D$@ (D ?D 0C$>Dt=C„: DT0D 0C¤BCT@C„AD$8C¢ ?D >Dd5D ACä2).Ð
/// </summary>Ð
public abstract class TechnologicalParameterBase : ReferenceBase, ITechnologicalParameterÐ
{Ð
    /// <summary>Ð
    /// A$=CTHCÔ8C’ :C´NDr =C  ACô@C 2CäGCÔ8C¢ 5CD8CÔ8Db 8Ct<CT@CT=C„O.Ð
    /// </summary>Ð
    public Guid? UnitOfMeasurementId { get; set; }Ð
Ð
    public virtual UnitOfMeasurement? UnitOfMeasurement { get; set; }Ð
Ð
    /// <summary>Ð
    /// B$8Cò 4C =CÔKDR ?C @C <CTBD 0 (CÔ0Cô@C„<CT@: Bt8D ;CâÂ !D$@Cä:C Â  D4;CT2Câ’í
    /// </summary>Ð
    public string DataType { get; set; } = "Numeric";Ð
Ð
    /// <summary>Ð
    /// B 0Ct@CTHCT=CÔKCR <CTBCä4D² @C AD GE BC
    /// </summary>Ð
    public CalculationMethod AllowedMethods { get; set; } = CalculationMethod.None;Ð
}
</file>
```

<file path="Promatis.Net.MES.Domain/TechnologicalParameter/TechnologicalParameterBaseValidator.cs">
```
using FluentValidation;Ð
using Promatis.Net.Domain;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
public abstract class TechnologicalParameterBaseValidator<T> : ReferenceBaseValidator<T>Ð
    where T : TechnologicalParameterBaseÐ
{Ð
    private static readonly string[] AllowedDataTypes = ["Numeric", "String", "Boolean",
"DateTime"];Ð
Ð
    protected TechnologicalParameterBaseValidator() : base()Ð
    {Ð
        // 1. A$0C´8CD0Dd8Dò 5CD8CÔ8DdK C„7CÄ5D 5CÔ8Dò „4CäAD$CCô=C Â BC : C¤0C¢ B r MD$> TechnologicalParame
        RuleFor(x => x.UnitOfMeasurement)Ð
            .NotNull()Ð
            .WithMessage("AT4C„=C„FC  8Ct<CT@CT=C„O CÔ5 CÄ>Cd5D" 1D´BDÂ çVÆÂâ"“½
Ð
        // 2. A$0C´8CD0Dd8Dò BC„?C  4C =CÔKD]
        RuleFor(x => x.DataType)Ð
            .NotEmpty()Ð
            .WithMessage("B$8Cò 4C =CÔKDR ?C @C <CTBD 0 Cä1Dô7C BCT;CT=.")Ð
            .Must(type => AllowedDataTypes.Contains(type))Ð
            .WithMessage($"AÔ5CD>CôCD BC„<D´9 D$8Cò 4C =CÔKDRâ  C 7D 5D„5CÔ=D´5 Ct=C GCT=C„O: {string.Join(",
    }Ð
}
```
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalParameterCalcMethod/
TechnologicalParameterCalcMethodBase.cs">
```
using Promatis.Net.Domain;Ð
using Promatis.Net.Domain.Interface;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
public abstract class TechnologicalParameterCalcMethodBase<TUnit, TOperation, TParameter> :
DomainObject, ISoftDeletableÐ
    where TUnit : UnitBaseÐ
    where TOperation : DomainObject, ITechnologicalOperationÐ
    where TParameter : TechnologicalParameterBaseÐ
{Ð
    public Guid UnitId { get; set; }Ð
Ð
    /// <summary>Ð
    /// Bd5DT>C$0Dò 5CD8CÔ8Dd0Ð
    /// </summary>Ð
    public virtual TUnit Unit { get; set; } = null!;Ð
Ð
    public Guid TechnologicalOperationId { get; set; }Ð
Ð
    /// <summary>Ð
    /// B$5DT=Cä;Cä3C„GCTAC¤0Dò >Cô5D 0Dd8Dý
    /// </summary>Ð
    public virtual TOperation TechnologicalOperation { get; set; } = null!;Ð
Ð
    public Guid TechnologicalParameterId { get; set; }Ð
Ð
    /// <summary>Ð
    /// B$5DT=Cä;Cä3C„GCTAC¤8C’ ?C @C <CTBD
    /// </summary>Ð
    public virtual TParameter TechnologicalParameter { get; set; } = null!;Ð
Ð
    /// <summary>Ð
    /// AÄ5D$>CB @C ADt5D$0Ð
    /// </summary>Ð
    public CalculationMethod CalculationMethod { get; set; } = CalculationMethod.None;Ð
Ð
    public DateTime? DeletedAt { get; set; }Ð
    public string? DeletedBy { get; set; }Ð
}
```
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalParameterCalcMethod/
TechnologicalParameterCalcMethodBaseValidator.cs">

```
using FluentValidation;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
public abstract class TechnologicalParameterCalcMethodBaseValidator<T, TUnit, TOperation,
TParameter> : DomainObjectValidator<T>Ð
    where T : TechnologicalParameterCalcMethodBase<TUnit, TOperation, TParameter>Ð
    where TUnit : UnitBaseÐ
    where TOperation : DomainObject, ITechnologicalOperationÐ
    where TParameter : TechnologicalParameterBaseÐ
{Ð
    protected TechnologicalParameterCalcMethodBaseValidator()Ð
    {Ð
        RuleFor(x => x.UnitId)Ð
            .NotEmpty().WithMessage("A„4CT=D$8DD8C¤0D$>D  FCTECä2Cä9 CT4C„=C„FD² >C OCt0D$5C´5CÒâ"“½
Ð
        RuleFor(x => x.TechnologicalOperationId)Ð
            .NotEmpty().WithMessage("A„4CT=D$8DD8C¤0D$>D  BCTECÔ>C´>C48Dt5D :Cä9 Cä?CT@C FC„8 Cä1Dô7C BCT;CT=
Ð
        RuleFor(x => x.TechnologicalParameterId)Ð
            .NotEmpty().WithMessage("A„4CT=D$8DD8C¤0D$>D  BCTECÔ>C´>C48Dt5D :Cä3Câ ?C @C <CTBD 0 Cä1Dô7C BCT;
Ð
        // Aô@Cä2CT@Dô5CÂÂ GD$> CÄ5D$>CB @C ADt5D$0 C KC² OC$=Câ 2D´1D 0CÒ ?Cä;DÄ7Cä2C BCT;CT<Ð
        RuleFor(x => x.CalculationMethod)Ð
            .NotEqual(CalculationMethod.None)Ð
            .WithMessage("AÔ5Cä1DT>CD8CÄ> C$KC @C BDÂ :Cä@D 5C¤BCÔKC’ <CTBCä4 D 0D GCTBC  ?C @C <CTBD >C"â"“½
    }Ð
}
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalParameterValue/TechnologicalParameterValueBase.cs">
using Promatis.Net.Domain;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
public abstract class TechnologicalParameterValueBase<TUnit, TParameter> : DomainObjectÐ
    where TUnit : UnitBaseÐ
    where TParameter : TechnologicalParameterBaseÐ
{Ð
    public Guid UnitId { get; set; }Ð
Ð
    /// <summary>Ð
    /// Bd5DT>C$0Dò 5CD8CÔ8Dd0 (C„AD$>Dt=C„: CD0CÔ=D´E)Ð
    /// </summary>Ð
    public virtual TUnit Unit { get; set; } = null!;Ð
Ð
    public Guid TechnologicalParameterId { get; set; }Ð
Ð
    /// <summary>Ð
    /// B$5DT=Cä;Cä3C„GCTAC¤8C’ ?C @C <CTBD
    /// </summary>Ð
    public virtual TParameter TechnologicalParameter { get; set; } = null!;Ð
Ð
    /// <summary>Ð
    /// B KD >CR ,$3D OCt=Cä5") Ct=C GCT=C„5, Cô5D 5CD0CÔ=Cä5 CäB SCADA/IoT C„;C, >Cô5D 0D$>D 0Ð
    /// </summary>Ð
    public string Value { get; set; } = string.Empty;Ð
Ð
    /// <summary>Ð
    /// B$>Dt=Cä5 C$@CT<Dò DC„:D 0Dd8C, ?C @C <CTBD 0 (Time-Series CÄ5D$:C •
    /// </summary>Ð
    public DateTime Date { get; set; } = DateTime.UtcNow;Ð
}
</file>

<file path="Promatis.Net.MES.Domain/TechnologicalParameterValue/
TechnologicalParameterValueBaseValidator.cs">
using FluentValidation;Ð
using Promatis.Net.Domain;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
public abstract class TechnologicalParameterValueBaseValidator<T, TUnit, TParameter> :
DomainObjectValidator<T>Ð
```

```
    where T : TechnologicalParameterValueBase<TUnit, TParameter>Ð
    where TUnit : UnitBaseÐ
    where TParameter : TechnologicalParameterBaseÐ
{Ð
    protected TechnologicalParameterValueBaseValidator()Ð
    {Ð
        RuleFor(x => x.UnitId)Ð
            .NotEmpty().WithMessage("A„4CT=D$8DD8C¤0D$>D  FCTECä2Cä9 CT4C„=C„FD² >C OCt0D$5C´5CÒâ"“½
Ð
        RuleFor(x => x.TechnologicalParameterId)Ð
            .NotEmpty().WithMessage("A„4CT=D$8DD8C¤0D$>D  BCTECÔ>C´>C48Dt5D :Cä3Câ ?C @C <CTBD 0 Cä1Dô7C BCT;
Ð
        RuleFor(x => x.Value)Ð
            .NotNull().WithMessage("At=C GCT=C„5 Cô0D 0CÄ5D$@C  =CR <Cä6CTB C KD$L null.")Ð
            // B 0Ct@CTHC 5CÂ ?D4AD$CDâ AD$@Cä:D2Â BC : C¤0C¢ 4C´O String-Cô0D 0CÄ5D$@Cä2 DÔBCâ <Cä6CTB C KD$
            // CÔ> D41C„@C 5CÂ AC´CDt0C”=D´5 Cô@Cä1CT;D² ?Câ :D 0Dô<Ð
            .Must(val => val == null || val.Trim() == val)Ð
            .WithMessage("At=C GCT=C„5 CÔ5 CÄ>Cd5D" =C GC„=C BDÄADò 8C´8 Ct0C¤0CÔGC„2C BDÄADò ?D >C 5C´0CÄ8."
Ð
        RuleFor(x => x.Date)Ð
            .Must(date => date <= DateTime.UtcNow.AddSeconds(5))Ð
            .WithMessage("A$@CT<Dò DC„:D 0Dd8C, ?C @C <CTBD 0 CÔ5 CÄ>Cd5D" 1D´BDÂ 2 C CCDCD”5CÂâ"“½
    }Ð
}
</file>

<file path="Promatis.Net.MES.Domain/UnitBase/UnitBase.cs">
using Promatis.Net.Domain;Ð
using Promatis.Net.Domain.Interface;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
/// <summary>Ð
/// A 1D BD 0C¤BCÔ0Dò 1C 7C  4C´O C$ACTE Cä@C40CÔ8Ct0Dd8Cä=CÔKDR 8 Cô@Cä8Ct2Cä4D BC$5CÔ=D´E CT4C„=C„F.Ð
/// </summary>Ð
public abstract class UnitBase : ReferenceTreeBase<UnitBase>, IAuditÐ
{Ð
    public UnitKind Kind { get; init; }Ð
    public required UnitType Type { get; set; }Ð
}
</file>

<file path="Promatis.Net.MES.Domain/UnitBase/UnitBaseValidator.cs">
using FluentValidation;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
/// <summary>Ð
/// A 1D BD 0C¤BCÔKC’ 2C ;C„4C BCä@ CD;Dò 1C 7Cä2Cä3Câ :C´0D AC  Væ—D& 6Rí
/// </summary>Ð
/// <typeparam name="T">B$8CòÂ =C AC´5CDCCT<D´9 CäB UnitBase.</typeparam>Ð
public abstract class UnitBaseValidator<T> : ReferenceBaseValidator<T>Ð
    where T : UnitBaseÐ
{Ð
    protected UnitBaseValidator() : base()Ð
    {Ð
        // Aô@Cä2CT@C¤0, DtBCâ 2 Type CÔ5 Cô@C„HC´> CôCD BCä5/CD5DD>C´BCÔ>CR 7CÔ0Dt5CÔ8CR DC´0C4>C-
        RuleFor(x => x.Type).NotEmpty();Ð
Ð
        // Aô@Cä2CT@C¤0 D >CäBC$5D$AD$2C„O A¤0D$5C4>D 8C, „¶–æB’ 8 B$8Cô0 (Type) CÔ0 CäACÔ>C$5 C 8D$>C$KDR <C
        RuleFor(x => x)Ð
            .Must(u => ((int)u.Kind & (int)u.Type) != 0)Ð
            .WithMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB C¤0D$5C4>D 8C,â"“½
    }Ð
}
</file>

<file path="Promatis.Net.MES.Domain/UnitBase/UnitHierarchyEngine.cs">
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
public static class UnitHierarchyEngineÐ
```

```
{Ð
    /// <summary>Ð
    /// A4;Cä1C ;DÄ=D´9 C„AD$>Dt=C„: Cô@C 2CDK: Cô@Cä2CT@Dô5D"Â 2C ;C„4CÔ> C´8 C$;Cä6CT=C„5 D 5C 5CÔ:C  2 D >
    /// A„ACô>C´LCtCCTBD O CÔ0CÄ5D BC$> C¤0C¢ 2 B #A  -D$@C„3C45D 0DRÂ BC : C, 2 UI-C$0C´8CD0Dd8C,í
    /// </summary>Ð
    public static bool IsHierarchyValid(UnitKind parentKind, UnitKind childKind)Ð
    {Ð
        // 1. B$5D <C„=C ;DÄ=D´9 D47CT; Position CÔ8C¤>C44C  =CR <Cä6CTB C KD$L D >CD8D$5C´5CÍ
        if (parentKind == UnitKind.Position) return false;Ð
Ð
        // 2. AD5Cô0D BC <CT=D" <Cä6CTB D >CD5D 6C BDÂ 2D Q, C¤@Cä<CR  ÷6—F–öí
        if (parentKind == UnitKind.Department) return childKind != UnitKind.Position;Ð
Ð
        // 3. B ?CTFC„0C´8Ct8D >C$0CÔ=D´5 Ct>CÔK C„7Cä;C„@Cä2C =D² 4D CC2 >D" 4D CC40, CÔ> CÄ>C4CD" ACä4CT@Cd
        if (parentKind is UnitKind.Production or UnitKind.Transport or UnitKind.Storage)Ð
        {Ð
            return childKind == parentKind || childKind == UnitKind.Position;Ð
        }Ð
Ð
        return true;Ð
    }Ð
Ð
    /// <summary>Ð
    /// A$>Ct2D 0D”0CTB D ?C„ACä: Aä Aä B  AÔ B´% D$8Cô>C" …Væ—EG— R’ 4C´O C$KCô0CD0DäICT3Câ ACô8D :C  2 UI,Ð
    /// CäACÔ>C$KC$0DôADÂ =C   C BCT3Cä@C„8 (UnitKind) D >CD8D$5C´LD :Cä3Câ CCt;C Â 2 C¤>D$>D KC’ <D² ACT9Dt0
    /// </summary>Ð
    public static IEnumerable<UnitType> GetAllowedChildTypes(UnitKind parentKind)Ð
    {Ð
        var allTypes = (UnitType[])Enum.GetValues(typeof(UnitType));Ð
        var allowedTypes = new List<UnitType>();Ð
Ð
        foreach (UnitType type in allTypes)Ð
        {Ð
            if (type == UnitType.None) continue;Ð
Ð
            // 1. B =C GC ;C  =C ECä4C„<, C¢ :C :Cä<D2 ?CäBCT=Dd8C ;DÄ=Cä<D2 ¶–æB >D$=CäAC„BD O DÔBCäB D$8Còí
            // B :C =C„@D45CÂ 2D 5 C¤0D$5C4>D 8C,Â GD$>C K Cô>CÔOD$L, C" :C :D4N CÄ0D :D2 ?Cä?C 4C 5D" MD$>D"
            foreach (UnitKind potentialChildKind in (UnitKind[])Enum.GetValues(typeof(UnitKind)))Ð
            {Ð
                // ATAC´8 D$8Cò ?Cä1C„BCä2Câ 2DT>CD8D" 2 DÔBD2 :C BCT3Cä@C„NÐ
                if (((int)potentialChildKind & (int)type) != 0)Ð
                {Ð
                    // A, 5D ;C, ?D 0C$8C´0 C„5D 0D EC„8 B  At AT(A .B" 2C´>Cd8D$L DÔBD2 :C BCT3Cä@C„N C" BCT
                    if (IsHierarchyValid(parentKind, potentialChildKind))Ð
                    {Ð
                        if (!allowedTypes.Contains(type))Ð
                        {Ð
                            allowedTypes.Add(type);Ð
                        }Ð
                    }Ð
                }Ð
            }Ð
        }Ð
Ð
        return allowedTypes;Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Domain/UnitOfMeasurement/UnitOfMeasurement.cs">
using Promatis.Net.Domain;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
/// <summary>Ð
/// AD>CÄ5CÔ=C O D CD”=CäAD$L CT4C„=C„FD² 8Ct<CT@CT=C„O (AÔ!A,’ 4C´O D$5DT=Cä;Cä3C„GCTAC¤8DR ?C @C <CTBD >C"
/// </summary>Ð
public class UnitOfMeasurement : ReferenceBaseÐ
{Ð
    // B 2Cä9D BC$0 Id, Code, Name, Description CÔ0D$8C$=Câ CCÔ0D ;CT4Cä2C =D² >D" &VfW&Væ6T& 6Rí
    // Code — C¤@C BC¤>CR >C >Ct=C GCT=C„5 (CÔ0Cô@C„<CT@, "°C", "CÄ<", "Cä1/CÄ8CÒ"’í
    // Name — Cô>C´=Cä5 CÔ0C„<CT=Cä2C =C„5 (CÔ0Cô@C„<CT@, "A4@C 4D4A Bd5C´LD 8Dò"Â $ C„;C´8CÄ5D$@").Ð
}
</file>

<file path="Promatis.Net.MES.Domain/UnitOfMeasurement/UnitOfMeasurementValidator.cs">
```

```
using FluentValidation;Ð
using Promatis.Net.Domain;Ð
Ð
namespace Promatis.Net.MES.Domain;Ð
Ð
public class UnitOfMeasurementValidator : ReferenceBaseValidator<UnitOfMeasurement>Ð
{Ð
    // Aô5D 5Cä?D 5CD5C´OCT< C 0Ct>C$CDâ @CT3D4;Dô@C¤C CÔ0 D 0D HC„@CT=CÔCDâ „@D4A/C´0D"Â FC„DD K, Cô@Cä1CT;D
    protected override string CodeRegexPattern => @"^[\p{L}\p{N}_\.\°\/\*\-\s\%]*$";Ð
Ð
    protected override string CodeValidationMessage => "A¤@C BC¤>CR >C >Ct=C GCT=C„5 CÄ>Cd5D" ACä4CT@Cd0D$L C
Ð
    public UnitOfMeasurementValidator() : base()Ð
    {Ð
        // Aô@C 2C„;Câ æ÷DV× G’ 8 Matches D46CR >D$@C 1CäBC ND" 2 C 0Ct>C$>CÂ :C´0D ACR ?Câ =C HCT<D2 =Cä2Cä<
Ð
        RuleFor(x => x.Code)Ð
            .NotEmpty().WithMessage("A¤@C BC¤>CR >C >Ct=C GCT=C„5 Cä1Dô7C BCT;DÄ=Câ 4C´O Ct0Cô>C´=CT=C„O.")Ð
            .MaximumLength(15).WithMessage("A¤@C BC¤>CR >C >Ct=C GCT=C„5 CÔ5 CD>C´6CÔ> Cô@CT2D´HC BDÂ  R AC„<
Ð
        RuleFor(x => x.Name)Ð
            .MaximumLength(100).WithMessage("Aô>C´=Cä5 CÔ0C„<CT=Cä2C =C„5 CT4C„=C„FD² 8Ct<CT@CT=C„O CÔ5 CD>C´
    }Ð
}
</file>

<file path="Promatis.Net.MES.Service/Promatis.Net.MES.Service.csproj">
<Project Sdk="Microsoft.NET.Sdk">Ð
Ð
  <PropertyGroup>Ð
    <TargetFramework>net10.0</TargetFramework>Ð
    <ImplicitUsings>enable</ImplicitUsings>Ð
    <Nullable>enable</Nullable>Ð
  </PropertyGroup>Ð
Ð
  <ItemGroup>Ð
    <ProjectReference Include="..\..\Core\Service\Promatis.Net.Service.csproj" />Ð
    <ProjectReference Include="..\Promatis.Net.MES.Data\Promatis.Net.MES.Data.csproj" />Ð
  </ItemGroup>Ð
Ð
</Project>
</file>

<file path="Promatis.Net.MES.Service/TechnologicalOperationParameterService/
ITechnologicalOperationParameterService.cs">
using Promatis.Net.Domain;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
public interface ITechnologicalOperationParameterService<T, TOperation, TParameter> :
IBaseService<T, Guid>Ð
    where T : TechnologicalOperationParameterBase<TOperation, TParameter>Ð
    where TOperation : DomainObject, ITechnologicalOperationÐ
    where TParameter : TechnologicalParameterBaseÐ
{Ð
    /// <summary>Ð
    /// Aô>C´CDt8D$L C$ACR ?C @C <CTBD K, Cô@C„2Dô7C =CÔKCR : C¤>CÔ:D 5D$=Cä9 Cä?CT@C FC„8Ð
    /// </summary>Ð
    Task<List<T>> GetByOperationIdAsync(Guid operationId);Ð
Ð
    /// <summary>Ð
    /// Aô>C´CDt8D$L D$>C´LC¤> Cä1Dô7C BCT;DÄ=D´5 Cô0D 0CÄ5D$@D² 4C´O Cä?CT@C FC„8Ð
    /// </summary>Ð
    Task<List<T>> GetRequiredParametersByOperationIdAsync(Guid operationId);Ð
}
</file>

<file path="Promatis.Net.MES.Service/TechnologicalOperationParameterService/
TechnologicalOperationParameterService.cs">
using Microsoft.EntityFrameworkCore;Ð
using Microsoft.EntityFrameworkCore.Internal;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.MES.Domain;Ð
```

```
using Promatis.Net.MES.Domain.Interface;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
public abstract class TechnologicalOperationParameterService<T, TOperation, TParameter, TContext>(Ð
    IDbContextFactory<TContext> contextFactory)Ð
    : BaseService<T, Guid, TContext>(contextFactory), ITechnologicalOperationParameterService<T,
TOperation, TParameter>Ð
    where T : TechnologicalOperationParameterBase<TOperation, TParameter>Ð
    where TOperation : DomainObject, ITechnologicalOperationÐ
    where TParameter : TechnologicalParameterBaseÐ
    where TContext : DbContextÐ
{Ð
    public async Task<List<T>> GetByOperationIdAsync(Guid operationId)Ð
    {Ð
        await using TContext context = await ContextFactory.CreateDbContextAsync();Ð
Ð
        return await context.Set<T>()Ð
            .Where(x => x.OperationId == operationId)Ð
            .Include(x => x.Parameter) // B @C 7D2 ?Cä4D$OC48C$0CT< D ?D 0C$>Dt=C„: Cô0D 0CÄ5D$@Cä2 CD;Dò T•
            .AsNoTracking()Ð
            .ToListAsync();Ð
    }Ð
Ð
    public async Task<List<T>> GetRequiredParametersByOperationIdAsync(Guid operationId)Ð
    {Ð
        await using TContext context = await ContextFactory.CreateDbContextAsync();Ð
Ð
        return await context.Set<T>()Ð
            .Where(x => x.OperationId == operationId && x.IsRequired)Ð
            .Include(x => x.Parameter)Ð
            .AsNoTracking()Ð
            .ToListAsync();Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Service/TechnologicalOperationService/
ITechnologicalOperationService.cs">
using Promatis.Net.Domain;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
public interface ITechnologicalOperationService<T, TLink> : ITreeService<T>Ð
    where T : ReferenceTreeBase<T>, ITechnologicalOperation,
Promatis.Net.Domain.Interface.IDomainObjectHasKey<Guid>Ð
    where TLink : classÐ
{Ð
    /// <summary>Ð
    /// Aô>C´CDt8D$L D ?C„ACä: D 0Ct@CTHCT=CÔ>C4> Cä1Cä@D44Cä2C =C„O (Dä=C„BCä2) CD;Dò :Cä=C¤@CTBCÔ>C’ BCTECÔ
    /// </summary>Ð
    Task<List<UnitBase>> GetAllowedUnitsAsync(Guid operationId);Ð
}
</file>

<file path="Promatis.Net.MES.Service/TechnologicalOperationService/
TechnologicalOperationService.cs">
using Microsoft.EntityFrameworkCore;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.Domain.Interface;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
public abstract class TechnologicalOperationService<T, TLink, TContext>(IDbContextFactory<TContext>
contextFactory)Ð
    : ReferenceTreeService<T, TContext>(contextFactory), ITechnologicalOperationService<T, TLink>Ð
    where T : ReferenceTreeBase<T>, ITechnologicalOperation, IDomainObjectHasKey<Guid>, new()Ð
    where TLink : TechnologicalOperationUnitBase<T>Ð
    where TContext : DbContextÐ
```

```
{Ð
    public async Task<List<UnitBase>> GetAllowedUnitsAsync(Guid operationId)Ð
    {Ð
        await using TContext context = await ContextFactory.CreateDbContextAsync();Ð
Ð
        // LINQ-Ct0Cô@CäA CäBD 0C 0D$KC$0CTB C„4CT0C´LCÔ>:Ð
        // A ;C 3Cä4C @Dò BCä<D2Â GD$> C" FV6†æöÆöv–6 Ä÷ W& F–öåVæ—D& 6R AC$>C”AD$2Câ Væ—B
        // C„<CT5D" AD$@Cä3C„9 D$8Cò Væ—D& 6RÂ Tb 6÷&R ?CäAD$@Cä8D" GC„AD$KC’ 5 ÂÔ¤ô”â 0C$BCä<C BC„GCTAC¤8.Ð
        return await context.Set<TLink>()Ð
            .Where(x => x.OperationId == operationId)Ð
            .Select(x => x.Unit)Ð
            .AsNoTracking()Ð
            .ToListAsync();Ð
    }Ð
Ð
    /// <summary>Ð
    /// AäAD$0C$;Dô5CÂ <CTBCä4 C 1D BD 0C¤BCÔKCÂ =C  BCT:D4ICT< D4@Cä2CÔ5. A¤>CÔ:D 5D$=D´9 Cô@C„:C´0CD=Cä9 D
    /// D$5DT?D >Dd5D ACä2 D 0CÂ ?CT@CT>Cô@CT4CT;C„B CT3Câ 4C´O D >Ct4C =C„O D 2Cä8DR ?Cä;C„<Cä@DD=D´E Cä?CT@
    /// </summary>Ð
    public abstract override Task<T> CreateChildTemplateAsync(T parent);Ð
}
</file>

<file path="Promatis.Net.MES.Service/TechnologicalParameterCalcMethodService/
ITechnologicalParameterCalcMethodService.cs">
using Promatis.Net.Domain;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
public interface ITechnologicalParameterCalcMethodService<T, TUnit, TOperation, TParameter> :
IBaseService<T, Guid>Ð
    where T : TechnologicalParameterCalcMethodBase<TUnit, TOperation, TParameter>Ð
    where TUnit : UnitBaseÐ
    where TOperation : DomainObject, ITechnologicalOperationÐ
    where TParameter : TechnologicalParameterBaseÐ
{Ð
    /// <summary>Ð
    /// AÔ0C”BC, :Cä=C¤@CTBCÔCDâ 8CÔAD$@D4:Dd8Dâ @C ADt5D$0 CD;Dò BD >C”:Cƒ¢  C >D CCD>C$0CÔ8CR ²  Cô5D 0Dd8D
    /// </summary>Ð
    Task<T?> GetCalcMethodAsync(Guid unitId, Guid operationId, Guid parameterId);Ð
Ð
    /// <summary>Ð
    /// AÔ>C´CDt8D$L C$ACR <CTBCä4D² @C ADt5D$0 CD;Dò :Cä=C¤@CTBCÔ>C’ >Cô5D 0Dd8C, =C  >C >D CCD>C$0CÔ8C•
    /// </summary>Ð
    Task<List<T>> GetMethodsByOperationAndUnitAsync(Guid operationId, Guid unitId);Ð
}
</file>

<file path="Promatis.Net.MES.Service/TechnologicalParameterCalcMethodService/
TechnologicalParameterCalcMethodService.cs">
using Microsoft.EntityFrameworkCore;Ð
using Microsoft.EntityFrameworkCore.Internal;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
public abstract class TechnologicalParameterCalcMethodService<T, TUnit, TOperation, TParameter,
TContext>(Ð
    IDbContextFactory<TContext> contextFactory)Ð
    : BaseService<T, Guid, TContext>(contextFactory), ITechnologicalParameterCalcMethodService<T,
TUnit, TOperation, TParameter>Ð
    where T : TechnologicalParameterCalcMethodBase<TUnit, TOperation, TParameter>Ð
    where TUnit : UnitBaseÐ
    where TOperation : DomainObject, ITechnologicalOperationÐ
    where TParameter : TechnologicalParameterBaseÐ
    where TContext : DbContextÐ
{Ð
    public async Task<T?> GetCalcMethodAsync(Guid unitId, Guid operationId, Guid parameterId)Ð
    {Ð
        await using TContext context = await ContextFactory.CreateDbContextAsync();Ð
```

```
Ð
        return await context.Set<T>()Ð
            .FirstOrDefaultAsync(x => x.UnitId == unitId &&Ð
                                      x.TechnologicalOperationId == operationId &&Ð
                                      x.TechnologicalParameterId == parameterId);Ð
    }Ð
Ð
    public async Task<List<T>> GetMethodsByOperationAndUnitAsync(Guid operationId, Guid unitId)Ð
    {Ð
        await using TContext context = await ContextFactory.CreateDbContextAsync();Ð
Ð
        return await context.Set<T>()Ð
            .Where(x => x.TechnologicalOperationId == operationId && x.UnitId == unitId)Ð
            .Include(x => x.TechnologicalParameter) // Aô>CD3D CCd0CT< D 2Dô7C =CÔKC’ ?C @C <CTBD
            .AsNoTracking()Ð
            .ToListAsync();Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Service/TechnologicalParameterService/
ITechnologicalParameterService.cs">
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
public interface ITechnologicalParameterService<T> : IReferenceService<T>Ð
    where T : TechnologicalParameterBaseÐ
{Ð
    /// <summary>Ð
    /// Aô>C´CDt8D$L Cô0D 0CÄ5D$@D²Â >D$DC„;DÄBD >C$0CÔ=D´5 Cô> D$8CôC CD0CÔ=D´E (Numeric, String, Boolean, D
    /// </summary>Ð
    Task<List<T>> GetByDataTypeAsync(string dataType);Ð
}
</file>

<file path="Promatis.Net.MES.Service/TechnologicalParameterService/
TechnologicalParameterService.cs">
using Microsoft.EntityFrameworkCore;Ð
using Microsoft.EntityFrameworkCore.Internal;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
public abstract class TechnologicalParameterService<T, TContext>(IDbContextFactory<TContext>
contextFactory)Ð
    : ReferenceService<T, TContext>(contextFactory), ITechnologicalParameterService<T>Ð
    where T : TechnologicalParameterBaseÐ
    where TContext : DbContextÐ
{Ð
    public async Task<List<T>> GetByDataTypeAsync(string dataType)Ð
    {Ð
        await using TContext context = await ContextFactory.CreateDbContextAsync();Ð
Ð
        return await context.Set<T>()Ð
            .Where(x => x.DataType == dataType)Ð
            .AsNoTracking()Ð
            .ToListAsync();Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Service/UnitOfMeasurementService/IUnitOfMeasurementService.cs">
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
/// <summary>Ð
/// A¤>CÔBD 0C¤B D 5D 2C„ACÔ>C4> D ;CäO CD;Dò CCô@C 2C´5CÔ8Dò ACô@C 2CäGCÔ8C¤>CÂ 5CD8CÔ8Db 8Ct<CT@CT=C„O.Ð
/// </summary>Ð
public interface IUnitOfMeasurementService : IReferenceService<UnitOfMeasurement>Ð
{Ð
}
```

```
</file>

<file path="Promatis.Net.MES.Service/UnitOfMeasurementService/UnitOfMeasurementService.cs">
using Microsoft.EntityFrameworkCore;Ð
using Promatis.Net.MES.Data;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
/// <summary>Ð
/// B 5D 2C„A D4?D 0C$;CT=C„O CÔ>D <C BC„2CÔ>-D ?D 0C$>Dt=Cä9 C„=DD>D <C FC„5C’ 5CD8CÔ8Db 8Ct<CT@CT=C„O.Ð
/// </summary>Ð
public class UnitOfMeasurementService(IDbContextFactory<MesApplicationDbContext> contextFactory)Ð
    : ReferenceService<UnitOfMeasurement, MesApplicationDbContext>(contextFactory),
IUnitOfMeasurementServiceÐ
{Ð
}
</file>

<file path="Promatis.Net.MES.Service/UnitService/IUnitBaseService.cs">
using Microsoft.EntityFrameworkCore;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.Domain.Interface;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
public interface IUnitBaseService<TContext> : IReferenceTreeService<UnitBase, TContext> // <- A„!Aô A  A´ AÔ
D$8Cô0 C" @Cä4C„BCT;DÄAC¤8C’ 8CÔBCT@DD5C”A!Ð
    where TContext : DbContextÐ
{Ð
    Task<List<UnitBase>> GetByKindAsync(UnitKind kind);Ð
    Task<List<UnitBase>> GetByTypeAsync(UnitType type);Ð
    Task<List<UnitBase>> GetByKindAndTypeAsync(UnitKind kind, UnitType type);Ð
}
</file>

<file path="Promatis.Net.MES.Service/UnitService/UnitBaseService.cs">
using Microsoft.EntityFrameworkCore;Ð
using Promatis.Net.Domain;Ð
using Promatis.Net.Domain.Interface;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Domain.Interface;Ð
using Promatis.Net.Service;Ð
Ð
namespace Promatis.Net.MES.Service;Ð
Ð
public abstract class UnitBaseService<TContext>(IDbContextFactory<TContext> contextFactory)Ð
    : ReferenceTreeService<UnitBase, TContext>(contextFactory), IUnitBaseService<TContext>Ð
    where TContext : DbContextÐ
{Ð
    // ========================================================================Ð
    // Aô A "BD B  AT AÔ+A’  A !B$ A  B$ B´  BT#A¢ „ AT A´ At#AT"B / AÔ  B  Aä  AÔ Ad )Ð
    // ========================================================================Ð
Ð
    /// <summary>Ð
    /// A¤>CÔBD 0C¤B DD0C @C„:C,â !C <C  @CT0C´8Ct0Dd8Dò 1D44CTB CÔ0Cô8D 0CÔ0 CÔ0 D ;Cä9 CÔ8Cd5 Ð
    /// C" :Cä=C¤@CTBCÔ>CÂ ACT@C$8D 5, C¤>D$>D KC’ 2C„4C„B Cô>C´8CÄ>D DCÔKCR :C´0D AD² „FW  'FÖVçEVæ—B 8 CD@.
    /// </summary>Ð
    public abstract override Task<UnitBase> CreateChildTemplateAsync(UnitBase parent);Ð
Ð
    // ========================================================================Ð
    // Bt B "B´  AÄ B$ AD+ A$+A  B  A,  A  AÔ+BR „!C$>C”AD$2C  ¶–æB 8 Type D$5Cô5D L C$8CD=D² 8CD5C ;DÄ=Câ•
    // ========================================================================Ð
Ð
    public async Task<List<UnitBase>> GetByKindAsync(UnitKind kind)Ð
    {Ð
        await using TContext context = await ContextFactory.CreateDbContextAsync();Ð
        return await context.Set<UnitBase>()Ð
            .AsNoTracking()Ð
            .Where(x => x.Kind == kind)Ð
            .ToListAsync();Ð
    }Ð
```

```
Đ
    public async Task<List<UnitBase>> GetByTypeAsync(UnitType type)Đ
    {Đ
        await using TContext context = await ContextFactory.CreateDbContextAsync();Đ
        // A„ACô>C´LCtCCT< Cô>C 8D$>C$>CR $ ", D$0C¢ :C : UnitType — DÔBCâ ´fÆ w5Ý
        return await context.Set<UnitBase>()Đ
            .AsNoTracking()Đ
            .Where(x => (x.Type & type) != 0)Đ
            .ToListAsync();Đ
    }Đ
Đ
    public async Task<List<UnitBase>> GetByKindAndTypeAsync(UnitKind kind, UnitType type)Đ
    {Đ
        await using TContext context = await ContextFactory.CreateDbContextAsync();Đ
        return await context.Set<UnitBase>()Đ
            .AsNoTracking()Đ
            .Where(x => x.Kind == kind && (x.Type & type) != 0)Đ
            .ToListAsync();Đ
    }Đ
}
</file>

<file path="Promatis.Net.MES.Tests/AssemblyInfo.cs">
// AäBC¤;DäGC 5D" ?C @C ;C´5C´LCÔ>CR 2D´?Cä;CÔ5CÔ8CR BCTAD$>C" 2 DÔBCä9 D 1Cä@C¤5Đ
[assembly: CollectionBehavior(DisableTestParallelization = true)]
</file>

<file path="Promatis.Net.MES.Tests/Promatis.Net.MES.Tests.csproj">
<Project Sdk="Microsoft.NET.Sdk">Đ
Đ
  <PropertyGroup>Đ
    <TargetFramework>net10.0</TargetFramework>Đ
    <ImplicitUsings>enable</ImplicitUsings>Đ
    <Nullable>enable</Nullable>Đ
    <IsPackable>false</IsPackable>Đ
  </PropertyGroup>Đ
Đ
  <ItemGroup>Đ
    <PackageReference Include="coverlet.collector" Version="10.0.0">Đ
      <PrivateAssets>all</PrivateAssets>Đ
      <IncludeAssets>runtime; build; native; contentfiles; analyzers; buildtransitive</
IncludeAssets>Đ
    </PackageReference>Đ
    <PackageReference Include="FluentAssertions" Version="8.10.0" />Đ
    <PackageReference Include="FluentValidation" Version="12.1.1" />Đ
    <PackageReference Include="Microsoft.EntityFrameworkCore.InMemory" Version="10.0.8" />Đ
    <PackageReference Include="Microsoft.Extensions.Configuration" Version="10.0.8" />Đ
    <PackageReference Include="Microsoft.NET.Test.Sdk" Version="18.5.1" />Đ
    <PackageReference Include="Moq" Version="4.20.72" />Đ
    <PackageReference Include="xunit.runner.visualstudio" Version="3.1.5">Đ
      <PrivateAssets>all</PrivateAssets>Đ
      <IncludeAssets>runtime; build; native; contentfiles; analyzers; buildtransitive</
IncludeAssets>Đ
    </PackageReference>Đ
    <PackageReference Include="xunit.v3" Version="3.2.2" />Đ
  </ItemGroup>Đ
Đ
  <ItemGroup>Đ
    <ProjectReference Include="..\Promatis.Net.Mes.Service\Promatis.Net.MES.Service.csproj" />Đ
  </ItemGroup>Đ
Đ
  <ItemGroup>Đ
    <Using Include="Xunit" />Đ
  </ItemGroup>Đ
Đ
</Project>
</file>

<file path="Promatis.Net.MES.Tests/Service/UnitServiceStandaloneTests.cs">
using FluentAssertions;Đ
using Microsoft.EntityFrameworkCore;Đ
using Microsoft.EntityFrameworkCore.ChangeTracking;Đ
using Microsoft.Extensions.Configuration;Đ
using Moq;Đ
using Promatis.Net.Data;Đ
using Promatis.Net.MES.Domain.Interface;Đ
```

```
using Promatis.Net.MES.Service;Ð
Ð
namespace Promatis.Net.MES.Tests;Ð
Ð
Ð
Ð
public class UnitServiceStandaloneTestsÐ
{Ð
    // 1. A„ACô>C´LCtCCT< C$5Ct4CR FW7DF$6öçFW‡M
    private readonly IDbContextFactory<TestDbContext> _factory;Ð
    private readonly DbContextOptions<TestDbContext> _options;Ð
Ð
    public class TestUnit : Domain.UnitBase { }Ð
Ð
    /// <summary>Ð
    /// B$5D BCä2D´9 D 5D 2C„A CD;Dò 2CT@C„DC„:C FC„8 C 8Ct=CTA-C$KC >D >C¢ Væ—D& 6R 2 CÄ>CDCC´LCÔKDR BCTAD$0
    /// A„!Aô A  A´ AÔ : B 8C4=C BD4@C  ?D 8C$5CD5CÔ0 C¢ AD$0CÔ4C @D$C UnitBaseService D  >CD=C„< generic-Cô0
    /// </summary>Ð
    public class TestUnitService(IDbContextFactory<TestDbContext> factory)Ð
        : UnitBaseService<TestDbContext>(factory)Ð
    {Ð
        /// <summary>Ð
        /// Aô@CäAD$5C”HC O Ct0C4;D4HC¤0 DD0C @C„:C, ?Cä;C„<Cä@DD8Ct<C  4C´O D4ACô5D„=Cä9 C¤>CÄ?C„;DôFC„8 C 1
        /// </summary>Ð
        public override Task<Domain.UnitBase> CreateChildTemplateAsync(Domain.UnitBase parent)Ð
        {Ð
            // A„!Aô A  A´ AÔ : A 5Ct>Cô0D =Cä5 CD8CÔ0CÄ8Dt5D :Cä5 D >Ct4C =C„5 C„=D BC =D 0 CÔ0D$8C$=Cä3Câ B
            var childMock = (Domain.UnitBase)Activator.CreateInstance(parent.GetType())!;Ð
Ð
            childMock.ParentId = parent.Id;Ð
            childMock.Parent = null;Ð
Ð
            return Task.FromResult(childMock);Ð
        }Ð
    }Ð
Ð
    public UnitServiceStandaloneTests()Ð
    {Ð
        // 2. Aä?Dd8C, 4Cä;Cd=D² 1D´BDÂ 4C´O TestDbContextÐ
        _options = new DbContextOptionsBuilder<TestDbContext>()Ð
            .UseInMemoryDatabase(Guid.NewGuid().ToString())Ð
            .Options;Ð
Ð
        // 3. AÄ>C¤0CT< DD0C @C„:D2 8CÄ5CÔ=Câ 4C´O TestDbContextÐ
        var factoryMock = new Mock<IDbContextFactory<TestDbContext>>();Ð
Ð
        factoryMock.Setup(f => f.CreateDbContextAsync(It.IsAny<CancellationToken>()))Ð
            .ReturnsAsync(() => new TestDbContext(_options)); // ReturnsAsync D4?D >D”0CTB Task.FromResultÐ
Ð
        _factory = factoryMock.Object;Ð
    }Ð
Ð
    // 4. A¤>CÔBCT:D B D$>Cd5 CD>C´6CT= Cô@C„=C„<C BDÂ ?D 0C$8C´LCÔKCR >CôFC„8Ð
    public class TestDbContext(DbContextOptions<TestDbContext> options)Ð
        : ApplicationDbContext(options, new ConfigurationBuilder().Build())Ð
    {Ð
        protected override void OnModelCreating(ModelBuilder modelBuilder)Ð
        {Ð
            base.OnModelCreating(modelBuilder);Ð
Ð
            modelBuilder.Entity<Domain.UnitBase>().UseTptMappingStrategy();Ð
            modelBuilder.Entity<TestUnit>().ToTable("TestUnits");Ð
        }Ð
Ð
        public override async Task<int> SaveChangesAsync(CancellationToken cancellationToken =
default)Ð
        {Ð
            IEnumerable<EntityEntry<Domain.UnitBase>> entries =
ChangeTracker.Entries<Domain.UnitBase>()Ð
                .Where(e => e.State is EntityState.Added or EntityState.Modified);Ð
Ð
            foreach (EntityEntry<Domain.UnitBase> entry in entries)Ð
            {Ð
                Domain.UnitBase unit = entry.Entity;Ð
                if (unit.ParentId.HasValue)Ð
                {Ð
```

```
                    // At0C4@D46C 5CÂ @Cä4C„BCT;Dò 4C´O Cô@Cä2CT@C¤8 (C" –äÖVÖ÷'' f–æB @C 1CäBC 5D" 1D´AD$@Câ
                    Domain.UnitBase? parent = Set<Domain.UnitBase>().Find(unit.ParentId.Value);Ð
                    if (parent != null)Ð
                    {Ð
                        // 1. Aô@C 2C„;Câ  ÷6—F–öâ „"CT@CÄ8CÔ0C´LCÔKC’ CCt5C²•
                        if (parent.Kind == UnitKind.Position)Ð
                            throw new InvalidOperationException("Position node cannot have
children.");Ð
Ð
                        // 2. Aô@C 2C„;Câ FW  'FÖVçB „ CR <Cä6CTB D >CD5D 6C BDÂ  ÷6—F–öâ•
                        if (parent.Kind == UnitKind.Department && unit.Kind == UnitKind.Position)Ð
                            throw new InvalidOperationException($"AÔ0D CD„5CÔ8CR 8CT@C @DT8Cƒ¢ >C JCT:D" :C B
'{parent.Kind}'.");Ð
Ð
                        // 3. Aô@C 2C„;Câ 8Ct>C´ODd8C, … &öGV7F–öâõ7F÷& vRõG& ç7 ÷'B =CR ACÄ5D„8C$0DäBD O)Ð
                        if (parent.Kind is UnitKind.Production or UnitKind.Storage or
UnitKind.Transport)Ð
                        {Ð
                            if (unit.Kind != parent.Kind && unit.Kind != UnitKind.Position)Ð
                                throw new InvalidOperationException($"AÔ0D CD„5CÔ8CR 8CT@C @DT8Cƒ¢ w·Væ—Bä¶–æ
'{parent.Kind}'.");Ð
                        }Ð
                    }Ð
                }Ð
            }Ð
Ð
            return await base.SaveChangesAsync(cancellationToken);Ð
        }Ð
    }Ð
Ð
Ð
Ð
    [Fact]Ð
    public async Task GetByKindAsync_Should_Return_Units_Matching_Complex_Mask()Ð
    {Ð
        // ArrangeÐ
        var service = new TestUnitService(_factory);Ð
Ð
        // BôGCT9C¤0 C$ECä4C„B C" <C AC¤C StorageÐ
        await service.AddAsync(new TestUnit { Id = Guid.NewGuid(), Name = "Cell_01", Kind =
UnitKind.Storage, Type = UnitType.Cell });Ð
        // B BC =Cä: C$ECä4C„B C" <C AC¤C ProductionÐ
        await service.AddAsync(new TestUnit { Id = Guid.NewGuid(), Name = "Lathe_01", Kind =
UnitKind.Production, Type = UnitType.MachineTool });Ð
Ð
        // ActÐ
        List<Domain.UnitBase> storageUnits = await service.GetByKindAsync(UnitKind.Storage);Ð
Ð
        // AssertÐ
        storageUnits.Should().ContainSingle(x => x.Name == "Cell_01");Ð
        storageUnits.Should().NotContain(x => x.Name == "Lathe_01");Ð
    }Ð
Ð
    [Fact]Ð
    public async Task GetByTypeAsync_Should_Handle_Specific_Flags()Ð
    {Ð
        // ArrangeÐ
        var service = new TestUnitService(_factory);Ð
        await service.AddAsync(new TestUnit { Id = Guid.NewGuid(), Name = "Crane_1", Kind =
UnitKind.Storage, Type = UnitType.Crane });Ð
        await service.AddAsync(new TestUnit { Id = Guid.NewGuid(), Name = "Vehicle_1", Kind =
UnitKind.Transport, Type = UnitType.Vehicle });Ð
Ð
        // ActÐ
        List<Domain.UnitBase> result = await service.GetByTypeAsync(UnitType.Crane);Ð
Ð
        // AssertÐ
        result.Should().ContainSingle(x => x.Name == "Crane_1");Ð
    }Ð
Ð
    [Fact]Ð
    public async Task MoveAsync_Should_Throw_When_Moving_Into_Position()Ð
    {Ð
        // ArrangeÐ
        var service = new TestUnitService(_factory);Ð
        var posId = Guid.NewGuid();Ð
```

```
        var machineId = Guid.NewGuid();Đ
Đ
        await service.AddAsync(new TestUnit { Id = posId, Name = "Terminal Point", Kind =
UnitKind.Position, Type = UnitType.Other });Đ
        await service.AddAsync(new TestUnit { Id = machineId, Name = "Mobile Machine", Kind =
UnitKind.Production, Type = UnitType.MachineTool });Đ
Đ
        // Act & AssertĐ
        Func<Task> act = async () => await service.MoveAsync(machineId, posId);Đ
Đ
        await act.Should().ThrowAsync<InvalidOperationException>()Đ
            .WithMessage("Position node cannot have children.");Đ
    }Đ
Đ
    [Fact]Đ
    public async Task MoveAsync_Should_Prevent_Circular_Dependencies()Đ
    {Đ
        // ArrangeĐ
        var service = new TestUnitService(_factory);Đ
        var idA = Guid.NewGuid();Đ
        var idB = Guid.NewGuid();Đ
Đ
        // AD>C 0C$;Dô5CÂ ¶–æB 8 Type, D$0C¢ :C : Cä=C, >C OCt0D$5C´LCÔK (required / init)Đ
        await service.AddAsync(new TestUnitĐ
        {Đ
            Id = idA,Đ
            Name = "Node A",Đ
            Kind = UnitKind.Department,Đ
            Type = UnitType.WorkshopĐ
        });Đ
Đ
        await service.AddAsync(new TestUnitĐ
        {Đ
            Id = idB,Đ
            Name = "Node B",Đ
            ParentId = idA,Đ
            Kind = UnitKind.Production,Đ
            Type = UnitType.SectionĐ
        });Đ
Đ
        // Act & AssertĐ
        Func<Task> act = async () => await service.MoveAsync(idA, idB);Đ
Đ
        await act.Should().ThrowAsync<InvalidOperationException>()Đ
            .WithMessage("*Bd8C¤;C„GCTAC¤0Dò 7C 2C„AC„<CäAD$L*");Đ
    }Đ
Đ
    [Fact]Đ
    public async Task GetFullTreeAsync_Should_Maintain_Hierarchy_And_References()Đ
    {Đ
        // ArrangeĐ
        var service = new TestUnitService(_factory);Đ
        var rootId = Guid.NewGuid();Đ
        var sectionId = Guid.NewGuid();Đ
Đ
        await service.AddAsync(new TestUnitĐ
        {Đ
            Id = rootId,Đ
            Name = "Dept",Đ
            Kind = UnitKind.Department,Đ
            Type = UnitType.WorkshopĐ
        });Đ
Đ
        await service.AddAsync(new TestUnitĐ
        {Đ
            Id = sectionId,Đ
            Name = "Section",Đ
            Kind = UnitKind.Production,Đ
            Type = UnitType.Section,Đ
            ParentId = rootIdĐ
        });Đ
Đ
        await service.AddAsync(new TestUnitĐ
        {Đ
            Id = Guid.NewGuid(),Đ
            Name = "Workstation",Đ
```

```
            Kind = UnitKind.Production,Đ
            Type = UnitType.Workstation,Đ
            ParentId = sectionIdĐ
        });Đ
Đ
        // ActĐ
        Domain.UnitBase? tree = await service.GetFullTreeAsync(rootId);Đ
Đ
        // AssertĐ
        tree.Should().NotBeNull();Đ
        tree!.Children.Should().HaveCount(1);Đ
        tree.Children.First().Children.Should().HaveCount(1);Đ
    }Đ
Đ
    [Fact]Đ
    public async Task MoveAsync_Department_Should_Not_Contain_Position()Đ
    {Đ
        // ArrangeĐ
        var service = new TestUnitService(_factory);Đ
        var deptId = Guid.NewGuid();Đ
        var positionId = Guid.NewGuid();Đ
Đ
        // 1. B >Ct4C 5CÂ  CT?C @D$0CÄ5CÔBĐ
        await service.AddAsync(new TestUnitĐ
        {Đ
            Id = deptId,Đ
            Name = "Main Dept",Đ
            Kind = UnitKind.Department,Đ
            Type = UnitType.WorkshopĐ
        });Đ
Đ
        // 2. B >Ct4C 5CÂ  Cä7C„FC„N (D >C4;C ACÔ> D$2Cä5C’ <C AC¤5, DÔBCâ <Cä6CTB C KD$L Cell)Đ
        await service.AddAsync(new TestUnitĐ
        {Đ
            Id = positionId,Đ
            Name = "Work Point 1",Đ
            Kind = UnitKind.Position,Đ
            Type = UnitType.CellĐ
        });Đ
Đ
        // ActĐ
        // AôKD$0CT<D O Cô5D 5CÄ5D BC„BDÂ  ÷6—F–öâ 2 DepartmentĐ
        Func<Task> act = async () => await service.MoveAsync(positionId, deptId);Đ
Đ
        // AssertĐ
        // Aä6C„4C 5CÂ >D„8C :D2 >D" Væ—D& 6T†–W& &6‡•G&–vvW-
        await act.Should().ThrowAsync<InvalidOperationException>()Đ
            .WithMessage("*C¤0D$5C4>D 8C, u ÷6—F–öâr =CR <Cä6CTB C KD$L C$;Cä6CT= C" tFW  'FÖVçBr¢"“½
    }Đ
Đ
    [Fact]Đ
    public async Task MoveAsync_Production_Should_Contain_Production_Subnode()Đ
    {Đ
        // ArrangeĐ
        var service = new TestUnitService(_factory);Đ
        var parentProdId = Guid.NewGuid();Đ
        var childProdId = Guid.NewGuid();Đ
Đ
        await service.AddAsync(new TestUnitĐ
        {Đ
            Id = parentProdId,Đ
            Name = "Production Area",Đ
            Kind = UnitKind.Production,Đ
            Type = UnitType.WorkshopĐ
        });Đ
Đ
        await service.AddAsync(new TestUnitĐ
        {Đ
            Id = childProdId,Đ
            Name = "Machine Node",Đ
            Kind = UnitKind.Production,Đ
            Type = UnitType.MachineToolĐ
        });Đ
Đ
        // ActĐ
        Func<Task> act = async () => await service.MoveAsync(childProdId, parentProdId);Đ
```

```
Ð
        // AssertÐ
        // At4CTADÂ 8D :C´NDt5CÔ8Dò 1D´BDÂ =CR 4Cä;Cd=CâÂ BC : C¤0C¢  &öGV7F–öâ <Cä6CTB D >CD5D 6C BDÂ  &öGV7
        await act.Should().NotThrowAsync();Ð
Ð
        Domain.UnitBase? updated = await service.GetByIdAsync(childProdId);Ð
        updated!.ParentId.Should().Be(parentProdId);Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Tests/UnitBase/AppDbContext.cs">
using Microsoft.EntityFrameworkCore;Ð
Ð
namespace Promatis.Net.MES.Tests.UnitBase;Ð
Ð
// B$5D BCä2D´9 C¤>CÔBCT:D B C$=D4BD 8 Cô@CäAD$@C =D BC$0 C„<CT= D$5D BCä2Ð
public class AppDbContext : DbContextÐ
{Ð
    public AppDbContext(DbContextOptions<AppDbContext> options) : base(options) { }Ð
Ð
    // A„ACô>C´LCtCCT< C 0Ct>C$KC’ BC„?, DtBCä1D² BD 8C43CT@ CÄ>C2 4CT;C BDÂ 6WCÅVæ—D& 6Sâ,•
    public DbSet<Domain.UnitBase> Units => Set<Domain.UnitBase>();Ð
Ð
    protected override void OnModelCreating(ModelBuilder modelBuilder)Ð
    {Ð
        // 1. AÔ0D BD 0C„2C 5CÂ E B „:C : C" >D =Cä2CÔ>CÂ ?D >CT:D$5)Ð
        modelBuilder.Entity<Domain.UnitBase>().UseTptMappingStrategy();Ð
Ð
        // 2. B4:C 7D´2C 5CÂÂ GD$> TestUnit — DÔBCâ =C AC´5CD=C„:Ð
        modelBuilder.Entity<TestUnit>().ToTable("TestUnits");Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.Tests/UnitBase/TestUnit.cs">
namespace Promatis.Net.MES.Tests.UnitBase;Ð
Ð
public class TestUnit : MES.Domain.UnitBase { }
</file>

<file path="Promatis.Net.MES.Tests/UnitBase/TestUnitValidator.cs">
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.Tests.UnitBase;Ð
Ð
public class TestUnitValidator : UnitBaseValidator<TestUnit> { }
</file>

<file path="Promatis.Net.MES.Tests/UnitBase/UnitBaseArchitectureTests.cs">
using FluentAssertions;Ð
using FluentValidation.TestHelper;Ð
using Microsoft.EntityFrameworkCore;Ð
using Promatis.Net.Data;Ð
using Promatis.Net.MES.Data;Ð
using Promatis.Net.MES.Domain.Interface;Ð
Ð
namespace Promatis.Net.MES.Tests.UnitBase;Ð
Ð
public class UnitBaseArchitectureTestsÐ
{Ð
    private readonly TestUnitValidator _validator = new();Ð
    private readonly DbContextOptions<AppDbContext> _options;Ð
Ð
    public UnitBaseArchitectureTests()Ð
    {Ð
        _options = new DbContextOptionsBuilder<AppDbContext>()Ð
            .UseInMemoryDatabase(Guid.NewGuid().ToString())Ð
            .Options;Ð
    }Ð
Ð
    private static EntityCancelEventArgs<MES.Domain.UnitBase> CreateArgs(MES.Domain.UnitBase
entity, DbContext context)Ð
    {Ð
        return new EntityCancelEventArgs<MES.Domain.UnitBase>(Ð
            entity,Ð
```

```
            EntityStateChangeEnum.Added,Ð
            new List<PropertyChangeInfo>(),Ð
            context);Ð
    }Ð
Ð
    #region A$0C´8CD0Dd8Dò „ C AC¤8 C, 1C 7Cä2D´5 Cô@C 2C„;C •
Ð
    [Theory]Ð
    [InlineData(UnitKind.Storage, UnitType.Cell, true)]      // BôGCT9C¤0 C" !C¤;C 4CR r  C-
    [InlineData(UnitKind.Storage, UnitType.Workshop, false)] // Bd5DR 2 B :C´0CD5 — AäHC„1C¤0 (C 8D" =CR ACä2
    [InlineData(UnitKind.Production, UnitType.Table, true)] // B BCä; C"  D >C„7C$>CDAD$2CR r  C-
    [InlineData(UnitKind.Position, UnitType.Other, true)]   // Aô@CäGCT5 C"  Cä7C„FC„8 — Aä:Ð
    [InlineData(UnitKind.Position, UnitType.Table, false)]  // B BCä; C"  Cä7C„FC„8 — AäHC„1C¤0 (D >C4;C ACÔ>
    public void Validator_Should_Check_Bitwise_Compatibility(UnitKind kind, UnitType type, bool
isValid)Ð
    {Ð
        var unit = new TestUnit { Kind = kind, Type = type, Name = "Test" };Ð
        TestValidationResult<TestUnit>? result = _validator.TestValidate(unit);Ð
Ð
        if (isValid)Ð
        {Ð
            result.ShouldNotHaveAnyValidationErrors();Ð
        }Ð
        elseÐ
        {Ð
            result.ShouldHaveValidationErrorFor(x => x)Ð
                  .WithErrorMessage("B$8Cò =CR ACä>D$2CTBD BC$CCTB C¤0D$5C4>D 8C,â"“½
        }Ð
    }Ð
Ð
    #endregionÐ
Ð
    #region B$@C„3C45D  „ D 0C$8C´0 C$;Cä6CT=CÔ>D BC, ¶–æB•
Ð
    [Fact]Ð
    public async Task Trigger_Should_Allow_Root_Units()Ð
    {Ð
        await using var context = new AppDbContext(_options);Ð
        var root = new TestUnit { ParentId = null, Kind = UnitKind.Department, Type =
UnitType.Other };Ð
Ð
        var trigger = new UnitBaseHierarchyTrigger();Ð
        EntityCancelEventArgs<Domain.UnitBase> args = CreateArgs(root, context);Ð
Ð
        await trigger.HandleAsync(args);Ð
Ð
        args.Cancel.Should().BeFalse();Ð
    }Ð
Ð
    [Fact]Ð
    public async Task Trigger_Should_Reject_Position_In_Department()Ð
    {Ð
        await using var context = new AppDbContext(_options);Ð
        var parentId = Guid.NewGuid();Ð
Ð
        context.Units.Add(new TestUnit { Id = parentId, Kind = UnitKind.Department, Type =
UnitType.Other, Name = "Dept" });Ð
        await context.SaveChangesAsync(TestContext.Current.CancellationToken);Ð
Ð
        var child = new TestUnit { ParentId = parentId, Kind = UnitKind.Position, Type =
UnitType.Other, Name = "Pos" };Ð
        var trigger = new UnitBaseHierarchyTrigger();Ð
        EntityCancelEventArgs<Domain.UnitBase> args = CreateArgs(child, context);Ð
Ð
        await trigger.HandleAsync(args);Ð
Ð
        args.Cancel.Should().BeTrue();Ð
        args.ErrorMessage.Should().Contain("CÔ5 CÄ>Cd5D" 1D´BDÂ 2C´>Cd5CÒ 2 'Dept' (Department)");Ð
    }Ð
Ð
    [Theory]Ð
    [InlineData(UnitKind.Production, UnitKind.Production, false)] // Bd5DR 2 Bd5DR r  C-
    [InlineData(UnitKind.Production, UnitKind.Transport, true)]   // B$@C =D ?Cä@D" 2 Bd5DR r  C ?D 5D”5CÔ> (
    [InlineData(UnitKind.Storage, UnitKind.Position, false)]      // Aô>Ct8Dd8Dò 2 B :C´0CB r  C-
    [InlineData(UnitKind.Position, UnitKind.Position, true)]      // Aô>Ct8Dd8Dò 2 Aô>Ct8Dd8Dâ r  C ?D 5D”5CÔ
    public async Task Trigger_Should_Enforce_Isolation(UnitKind pKind, UnitKind cKind, bool
shouldCancel)Ð
```

```
    {Ð
        await using var context = new AppDbContext(_options);Ð
        var parentId = Guid.NewGuid();Ð
Ð
        // B >Ct4C 5CÂ @Cä4C„BCT;Dò A Cô>CDECä4DôIC„< D$8Cô>CÍ
        context.Units.Add(new TestUnit { Id = parentId, Kind = pKind, Type = GetFirstType(pKind),
Name = "P" });Ð
        await context.SaveChangesAsync(TestContext.Current.CancellationToken);Ð
Ð
        var child = new TestUnit { ParentId = parentId, Kind = cKind, Type = GetFirstType(cKind),
Name = "C" };Ð
        EntityCancelEventArgs<Domain.UnitBase> args = CreateArgs(child, context);Ð
        await new UnitBaseHierarchyTrigger().HandleAsync(args);Ð
Ð
        args.Cancel.Should().Be(shouldCancel, $"Cô>D$>CÄC DtBCâ 2C´>Cd5CÔ8CR ¶4¶–æGÒ 2 {pKind} CD>C´6CÔ> Cô@C
    }Ð
Ð
    #endregionÐ
Ð
    private static UnitType GetFirstType(UnitKind kind)Ð
        => Enum.GetValues<UnitType>().First(t => t != UnitType.None && ((int)kind & (int)t) != 0);Ð
}
</file>

<file path="Promatis.Net.MES.UI/_Imports.razor">
@using Microsoft.AspNetCore.Components.FormsÐ
@using Microsoft.AspNetCore.Components.RoutingÐ
@using Microsoft.AspNetCore.Components.WebÐ
@using MudBlazorÐ
@using Promatis.Net.UI.Components.WorkspaceÐ
@using Promatis.Net.UI.Components.EditDialogÐ
@using Promatis.Net.UI.Components.GridÐ
@using Promatis.Net.UI.Components.ToolbarÐ
@using Promatis.Net.UI.Components.Tree
</file>

<file path="Promatis.Net.MES.UI/Pages/UnitOfMeasurements/Card/UnitOfMeasurementEditDialog.razor">
<EditDialog Title="@Title" Ð
            IsNew="@IsNew" Ð
            Model="@Model" Ð
            Validator="@Validator" Ð
            OnSaveAction="@OnSaveAction">Ð
    <Tabs>Ð
        <DialogTab Title="AäACÔ>C$=Cä5" Icon="@Icons.Material.Filled.Info">Ð
            <Content>Ð
                <div class="d-flex flex-column gap-4 pt-2">Ð
Ð
                    <MudTextField Label="A¤@C BC¤>CR >C >Ct=C GCT=C„5 (A¤>CB’"
                                  @bind-Value="Model.Code" Ð
                                  For="@(() => Model.Code)" Ð
                                  Margin="Margin.Dense" Ð
                                  Variant="Variant.Outlined" Ð
                                  Required="true"Ð
                                  HelperText="AÔ0Cô@C„<CT@: CÄ<, Cä1/CÄ8CÒÂ  2Â HD"" ói
                                  Ð
                    <MudTextField Label="Aô>C´=Cä5 CÔ0C„<CT=Cä2C =C„5" Ð
                                  @bind-Value="Model.Name" Ð
                                  For="@(() => Model.Name)" Ð
                                  Margin="Margin.Dense" Ð
                                  Variant="Variant.Outlined" Ð
                                  Required="true" />Ð
                                  Ð
                    <MudTextField Label="Aä?C„AC =C„5 / AÔ@C„<CTGC =C„5" Ð
                                  @bind-Value="Model.Description" Ð
                                  For="@(() => Model.Description)" Ð
                                  Margin="Margin.Dense" Ð
                                  Variant="Variant.Outlined" Ð
                                  Lines="3" />Ð
Ð
                </div>Ð
            </Content>Ð
        </DialogTab>Ð
    </Tabs>Ð
</EditDialog>
</file>
```

<file path="Promatis.Net.MES.UI/Pages/UnitOfMeasurements/Card/UnitOfMeasurementEditDialog.razor.cs">
using Microsoft.AspNetCore.Components;Ð
using Promatis.Net.MES.Domain;Ð
Ð
namespace Promatis.Net.MES.UI.Pages.UnitOfMeasurements.Card;Ð
Ð
public partial class UnitOfMeasurementEditDialog : ComponentBaseÐ
{Ð
    [Parameter] public required string Title { get; set; }Ð
    [Parameter] public required bool IsNew { get; set; }Ð
    [Parameter] public required UnitOfMeasurement Model { get; set; }Ð
    [Parameter] public required FluentValidation.IValidator Validator { get; set; }Ð
    [Parameter] public required Func<Task> OnSaveAction { get; set; }Ð
}
</file>

<file path="Promatis.Net.MES.UI/Pages/UnitOfMeasurements/UnitOfMeasurementPage.razor">
@page "/mes/mdm/units-of-measurement"Ð
Ð
@using Promatis.Net.MES.DomainÐ
Ð
@inject UnitOfMeasurementPageContext ContextÐ
Ð
<CascadingValue Value="@Context" IsFixed="true">Ð
    <WorkspacePage>Ð
Ð
        <TopContent>Ð
            <UiToolbar TEntity="UnitOfMeasurement"Ð
                       ActionContext="@Context"Ð
                       OnCreateTriggered="@Context.OnCreateActionAsync"Ð
                       OnEditTriggered="@Context.OnUpdateActionAsync"Ð
                       OnDeleteTriggered="@Context.OnDeleteActionAsync" />Ð
        </TopContent>Ð
Ð
        <BodyContent>Ð
            @* Bt8D BC O CD5C¤;C @C FC„O: GridPage C 2D$>CÄ0D$8Dt5D :C, 7C 1CT@CTB CD0CÔ=D´5 C„7 Context.Data
            <GridPage TEntity="UnitOfMeasurement">Ð
                <ColumnsContent>Ð
                    <PropertyColumn T="UnitOfMeasurement" TProperty="string" Property="x => x.Code"Ð
                                    Title="A¤>CB 5CD8CÔ8DdK C„7CÄ5D 5CÔ8Dò" †V FW%7G–ÆSÒ'v–GFƒ¢ S ƒ²" 6Æ 73
                    <PropertyColumn T="UnitOfMeasurement" TProperty="string" Property="x => x.Name"Ð
                                    Title="AÔ0C„<CT=Cä2C =C„5 CT4. C„7CÂâ" †V FW%7G–ÆSÒ'v–GFƒ¢ 3   ƒ²" ói
                    <PropertyColumn T="UnitOfMeasurement" TProperty="string" Property="x =>
x.Description"Ð
                                    Title="Aä?C„AC =C„5 / AÔ@C„<CTGC =C„5" />Ð
                </ColumnsContent>Ð
            </GridPage>Ð
        </BodyContent>Ð
Ð
    </WorkspacePage>Ð
</CascadingValue>
</file>

<file path="Promatis.Net.MES.UI/Pages/UnitOfMeasurements/UnitOfMeasurementPageContext.cs">
using MudBlazor;Ð
using Promatis.Net.Domain.Interface;Ð
using Promatis.Net.MES.Domain;Ð
using Promatis.Net.MES.Service;Ð
using Promatis.Net.MES.UI.Pages.UnitOfMeasurements.Card;Ð
using Promatis.Net.UI.Components.Grid;Ð
Ð
namespace Promatis.Net.MES.UI.Pages.UnitOfMeasurements;Ð
Ð
public class UnitOfMeasurementPageContext : GridActionContext<UnitOfMeasurement>Ð
{Ð
    private readonly IUnitOfMeasurementService _uomService;Ð
    private readonly IDialogService _dialogService;Ð
    private readonly IEntityCloner _entityCloner;Ð
Ð
    public UnitOfMeasurementPageContext(Ð
        IUnitOfMeasurementService uomService,Ð
        IDialogService dialogService,Ð
        IEntityCloner entityCloner)Ð
    {Ð
        _uomService = uomService ?? throw new ArgumentNullException(nameof(uomService));Ð
        _dialogService = dialogService ?? throw new ArgumentNullException(nameof(dialogService));Ð

```
        _entityCloner = entityCloner ?? throw new ArgumentNullException(nameof(entityCloner));Ð
Ð
        PageTitle = "B ?D 0C$>Dt=C„: CT4C„=C„F C„7CÄ5D 5CÔ8Dò#½
Ð
        // A$:C´NDt0CT< CÔ0D$8C$=D´9 Aä B2Ô@CT6C„< CD;Dò ?D 8C$OCt:C, : GridPageÐ
        DataBroker.ConfigureInMemoryMode();Ð
    }Ð
Ð
    /// <summary>Ð
    /// B$ Bt Bt Aä  AD A  A$ AT A„ : AÄ5D$>CB ?CT@C$8Dt=Cä3Câ =C ?Cä;CÔ5CÔ8Dò  At#-C¤MD„0 C @Cä:CT@C  4C =CÔ
    /// </summary>Ð
    public async Task InitializeInMemoryDataAsync()Ð
    {Ð
        tryÐ
        {Ð
            List<UnitOfMeasurement> data = await _uomService.GetAllAsync();Ð
            DataBroker.InMemoryItems = data;Ð
        }Ð
        catch (Exception ex)Ð
        {Ð
            Console.WriteLine($"[ERROR] AäHC„1C¤0 C„=C„FC„0C´8Ct0Dd8C,  At#-C¤MD„0 CT4C„=C„F C„7CÄ5D 5CÔ8Dó¢
        }Ð
    }Ð
Ð
    // ============================================================================Ð
    // A¤ AÄ AÔ B²  A  AT A,  AÔ!B$ B4 AT B$ A" „5%TB•
    // ============================================================================Ð
Ð
    public async Task OnCreateActionAsync()Ð
    {Ð
        var newUom = new UnitOfMeasurement();Ð
        await OpenEditDialogAsync<UnitOfMeasurementEditDialog>(Ð
            newUom,Ð
            "AD>C 0C$;CT=C„5 CT4C„=C„FD² 8Ct<CT@CT=C„O",Ð
            isNew: true,Ð
            saveDelegate: () => _uomService.AddAsync(newUom)Ð
        );Ð
    }Ð
Ð
    public async Task OnUpdateActionAsync(UnitOfMeasurement? row)Ð
    {Ð
        if (row == null) return;Ð
        UnitOfMeasurement targetClone = _entityCloner.CloneEntity(row);Ð
        await OpenEditDialogAsync<UnitOfMeasurementEditDialog>(Ð
            targetClone,Ð
            $"B 5CD0C¤BC„@Cä2C =C„5: {row.Code}",Ð
            isNew: false,Ð
            saveDelegate: () => _uomService.UpdateAsync(targetClone)Ð
        );Ð
    }Ð
Ð
    public async Task OnDeleteActionAsync()Ð
    {Ð
        if (SelectedItem == null) return;Ð
Ð
        bool? confirm = await _dialogService.ShowMessageBoxAsync(Ð
            "B44C ;CT=C„5 Ct0Cô8D 8",Ð
            $"A$K CD5C”AD$2C„BCT;DÄ=Câ ECäBC„BCR CCD0C´8D$L CT4C„=C„FD2 8Ct<CT@CT=C„O '{SelectedItem.Name}' (
            yesText: "B44C ;C„BDÂ"Â 6 æ6VÅFW‡C¢ $ D$<CT=C "“½
Ð
        if (confirm == true)Ð
        {Ð
            await _uomService.DeleteAsync(SelectedItem.Id);Ð
        }Ð
    }Ð
Ð
}
</file>

<file path="Promatis.Net.MES.UI/Promatis.Net.MES.UI.csproj">
<Project Sdk="Microsoft.NET.Sdk.Razor">Ð
Ð
  <PropertyGroup>Ð
    <TargetFramework>net10.0</TargetFramework>Ð
    <Nullable>enable</Nullable>Ð
    <ImplicitUsings>enable</ImplicitUsings>Ð
```

```
  </PropertyGroup>Ð
Ð
  <ItemGroup>Ð
    <Compile Remove="UnitBase\**" />Ð
    <Content Remove="UnitBase\**" />Ð
    <EmbeddedResource Remove="UnitBase\**" />Ð
    <None Remove="UnitBase\**" />Ð
  </ItemGroup>Ð
Ð
Ð
  <ItemGroup>Ð
    <SupportedPlatform Include="browser" />Ð
  </ItemGroup>Ð
Ð
  <ItemGroup>Ð
    <PackageReference Include="Microsoft.AspNetCore.Components.Web" Version="10.0.8" />Ð
    <PackageReference Include="MudBlazor" Version="9.4.0" />Ð
  </ItemGroup>Ð
Ð
  <ItemGroup>Ð
    <ProjectReference Include="..\..\Core\Configuration.Web\Promatis.Net.Configuration.Web.csproj" /
>Ð
    <ProjectReference Include="..\..\Core\Promatis.Net.UI\Promatis.Net.UI.csproj" />Ð
    <ProjectReference Include="..\Promatis.Net.MES.Service\Promatis.Net.MES.Service.csproj" />Ð
  </ItemGroup>Ð
Ð
</Project>
</file>

<file path="Promatis.Net.MES.UI/UiModule.cs">
using MudBlazor;Ð
using Promatis.Net.UI;Ð
Ð
namespace Promatis.Net.MES.UI;Ð
Ð
public class UiModule : IUiModuleÐ
{Ð
    public string Name => GetType().Assembly.GetName().Name ?? "Unknown.Module";Ð
Ð
    public IEnumerable<(string Title, string Href, string Icon, string? Group)> GetMenuItems()Ð
    {Ð
        return new List<(string Title, string Href, string Icon, string? Group)>Ð
        {Ð
            // Aô;CäAC¤8C’ ACô@C 2CäGCÔ8C¢ BCTECÔ>C´>C48Dt5D :C„E Cô0D 0CÄ5D$@Cä2Ð
            ("AT4C„=C„FD² 8Ct<CT@CT=C„O", "/mes/mdm/units-of-measurement", Icons.Material.Filled.SettingsInpu
        };Ð
    }Ð
}
</file>

<file path="Promatis.Net.MES.UI/wwwroot/exampleJsInterop.js">
// This is a JavaScript module that is loaded on demand. It can export any number ofÐ
// functions, and may import other JavaScript modules if required.Ð
Ð
export function showPrompt(message) {Ð
  return prompt(message, 'Type anything here');Ð
}
</file>

</files>
```